



Analysing community data

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bit.ly/SSoQE



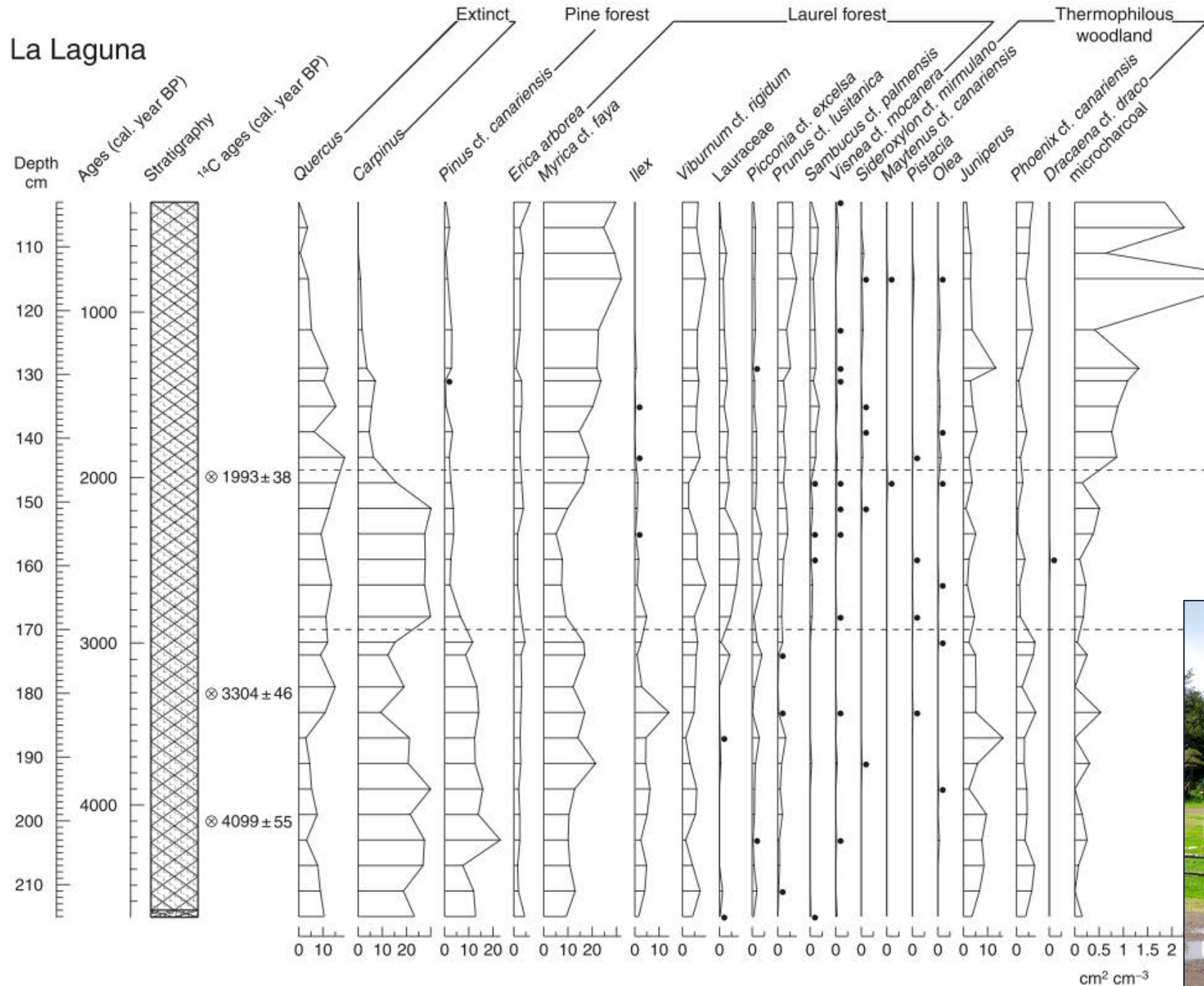
Aims

1. Understand the structure of **species community data**
2. Be able to interpret **Ordination** analyses
3. Be able to **build a simple Ordination** in R.
4. Know **resources** for future analyses of community data

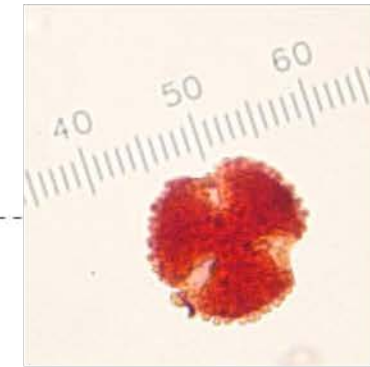
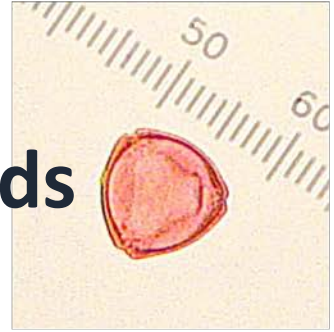


Tenerife, 3718m, Canary Islands, from Christoph Hormann, imagico.de

La Laguna

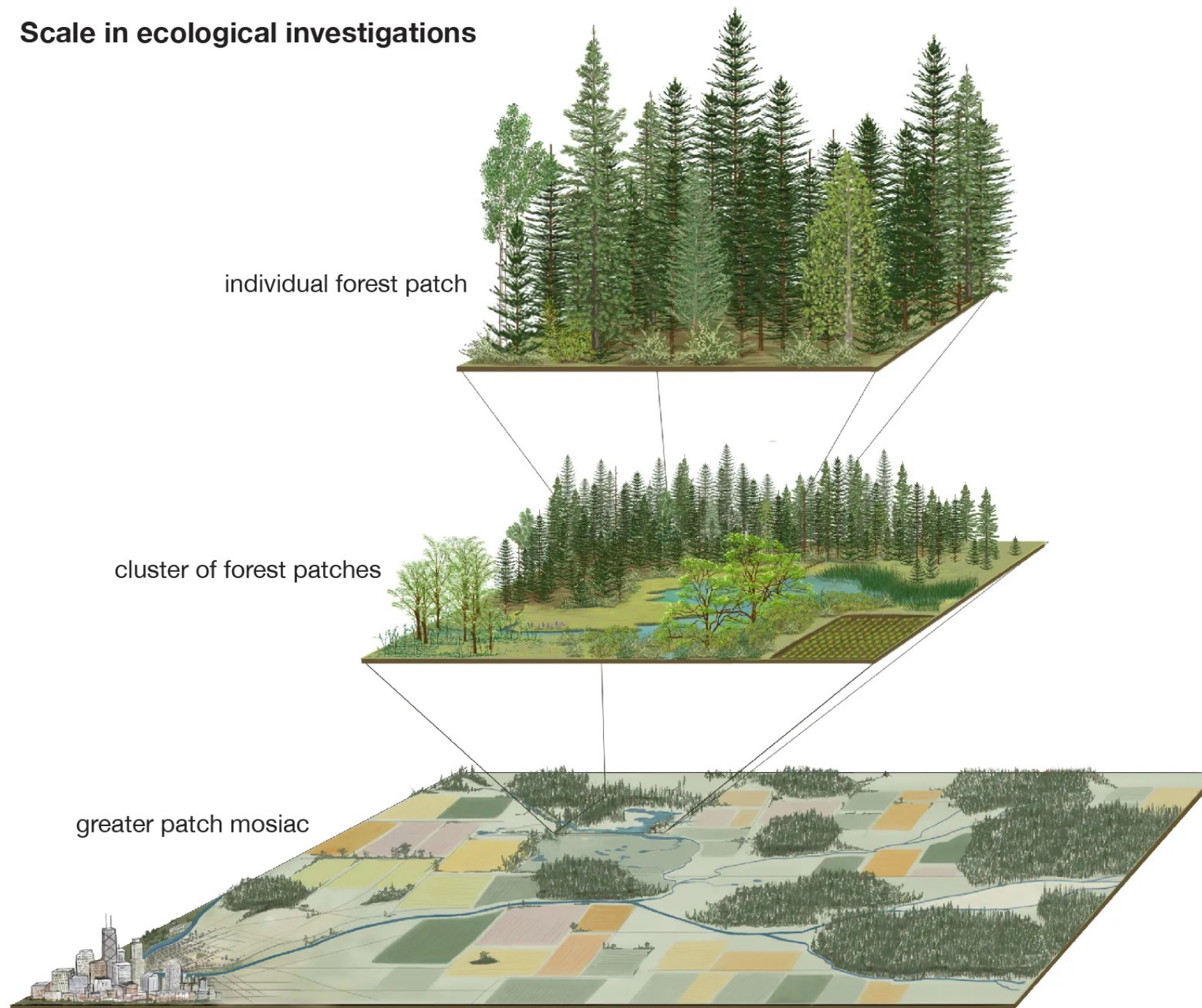


Tenerife, Canary Islands

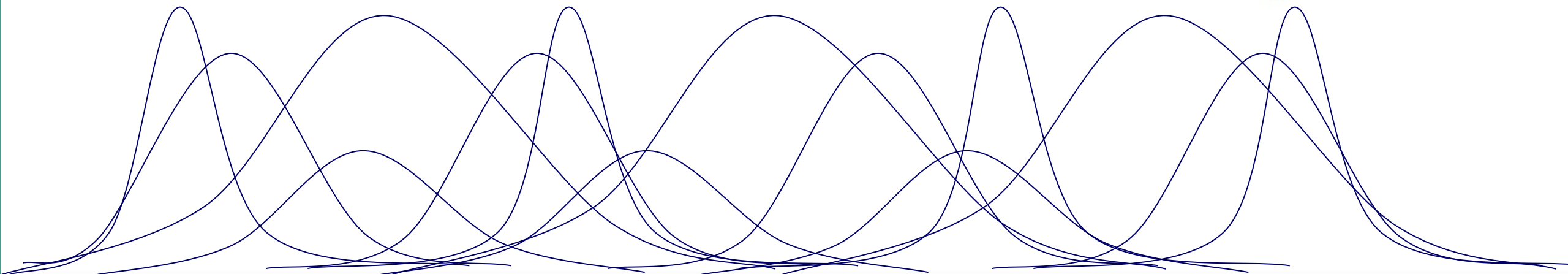
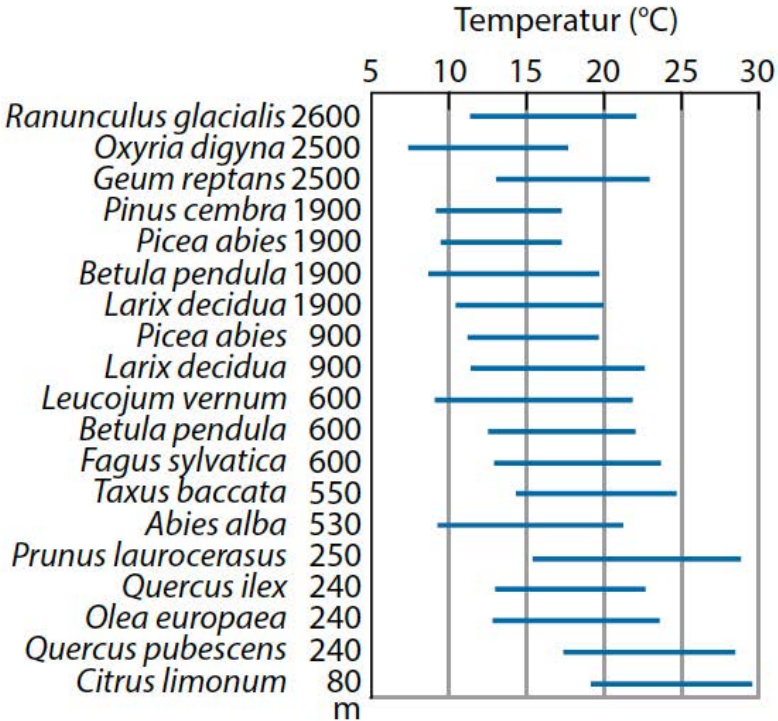




Scale in ecological investigations

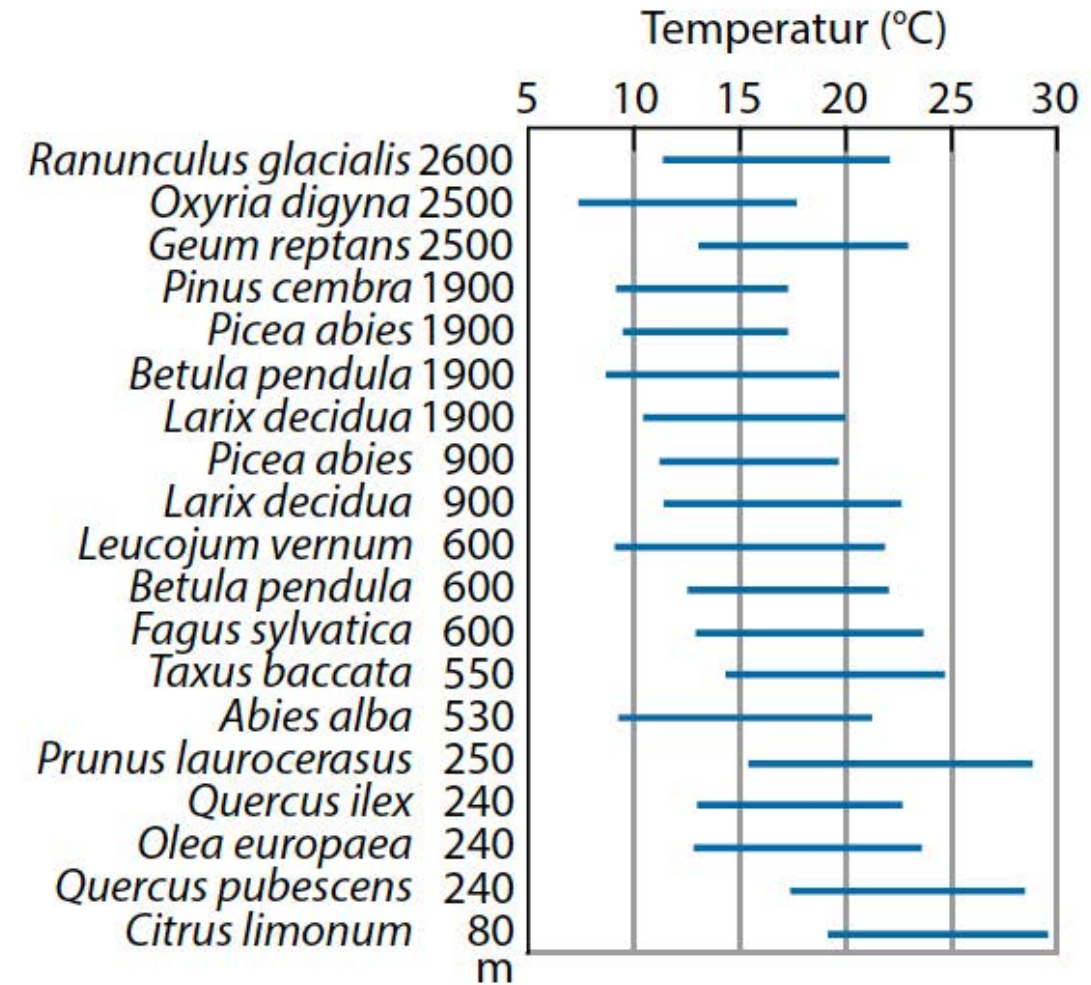
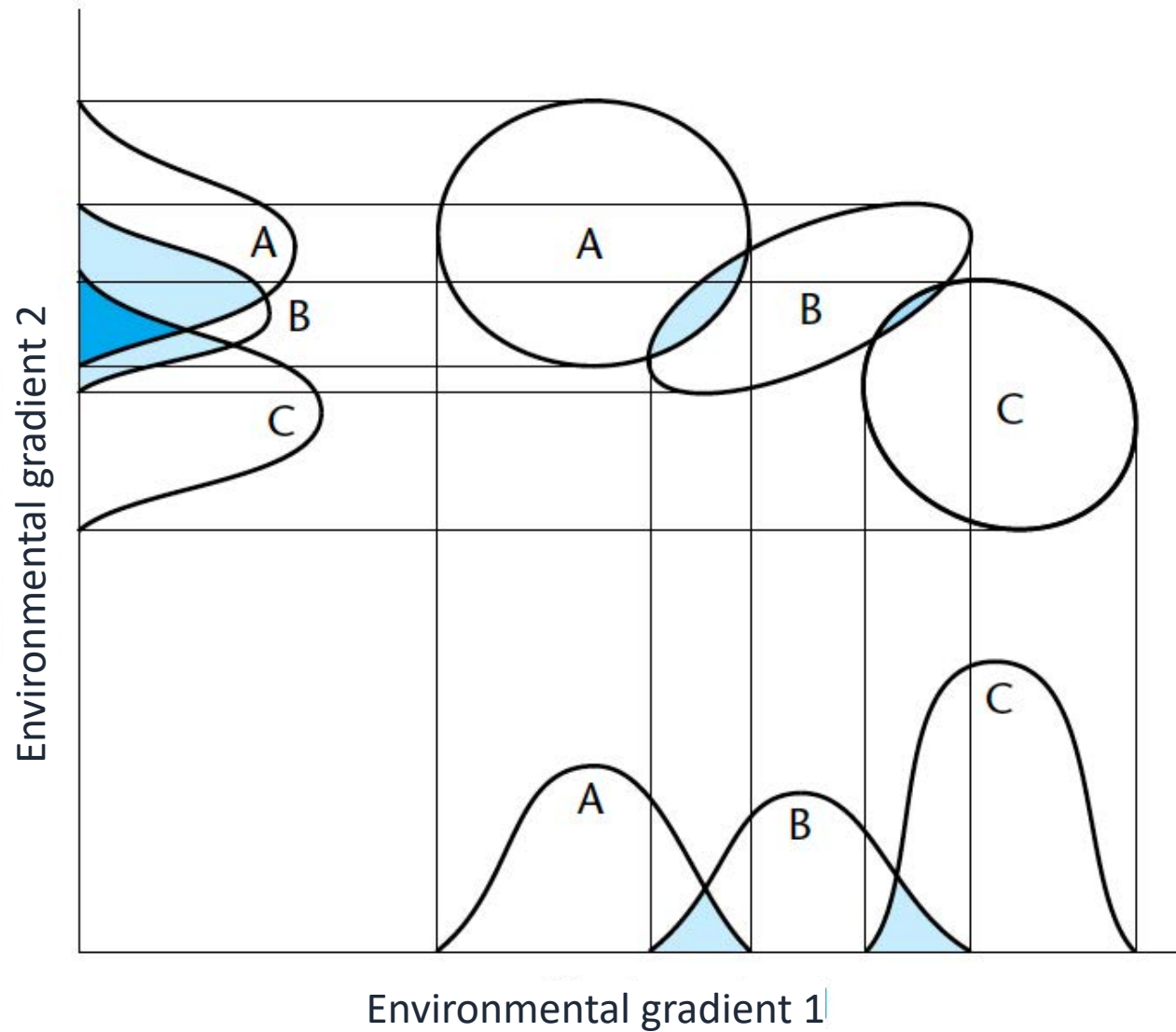


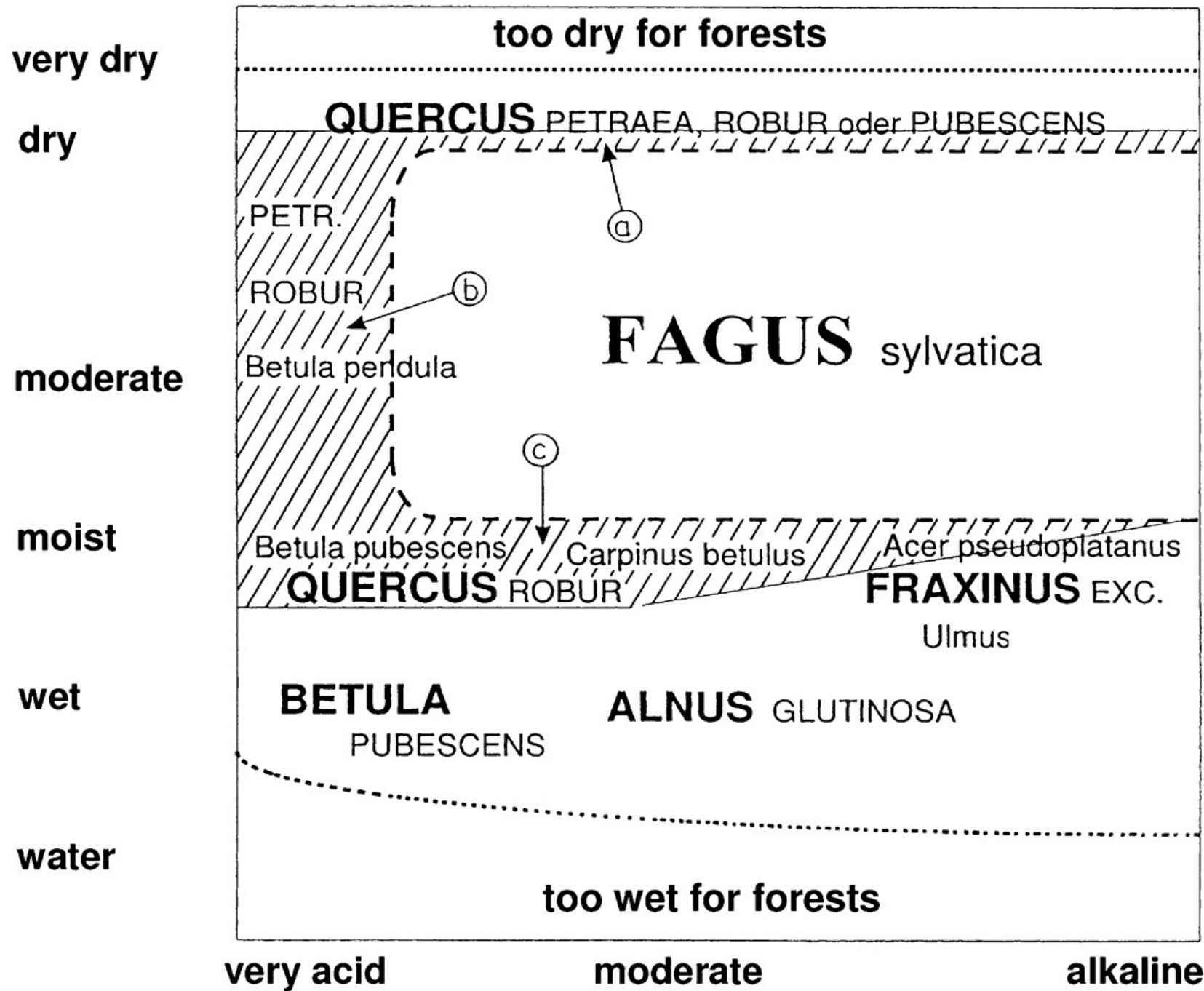
Environmental niche

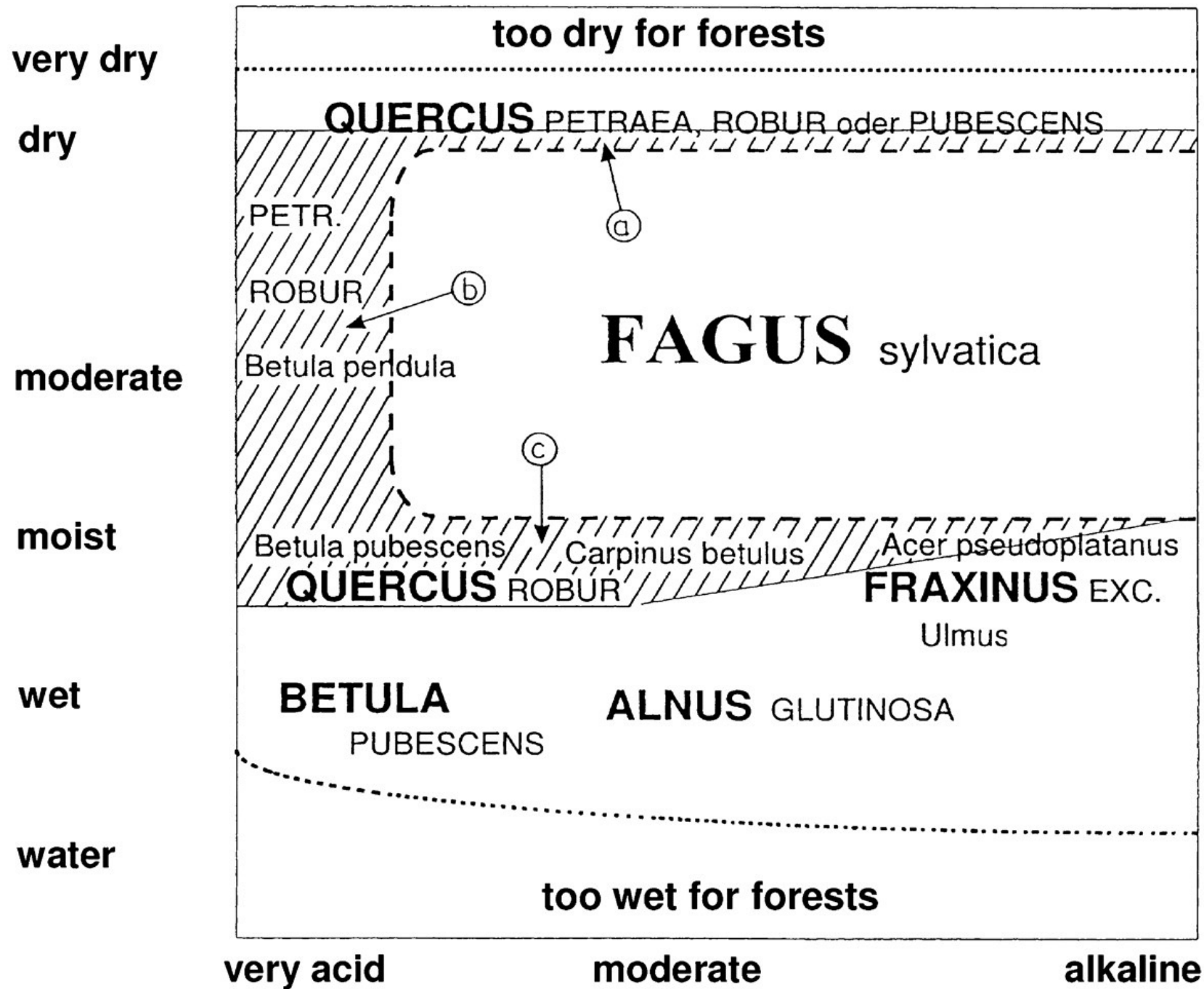


Temperature

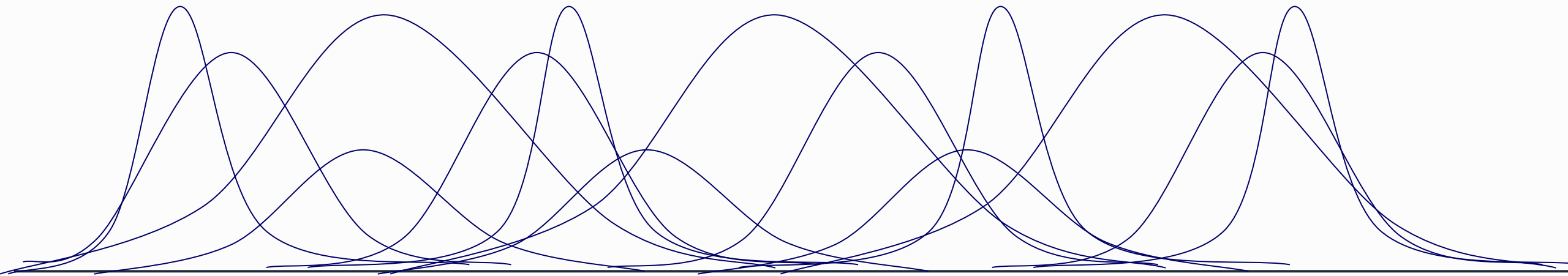
Environmental niche





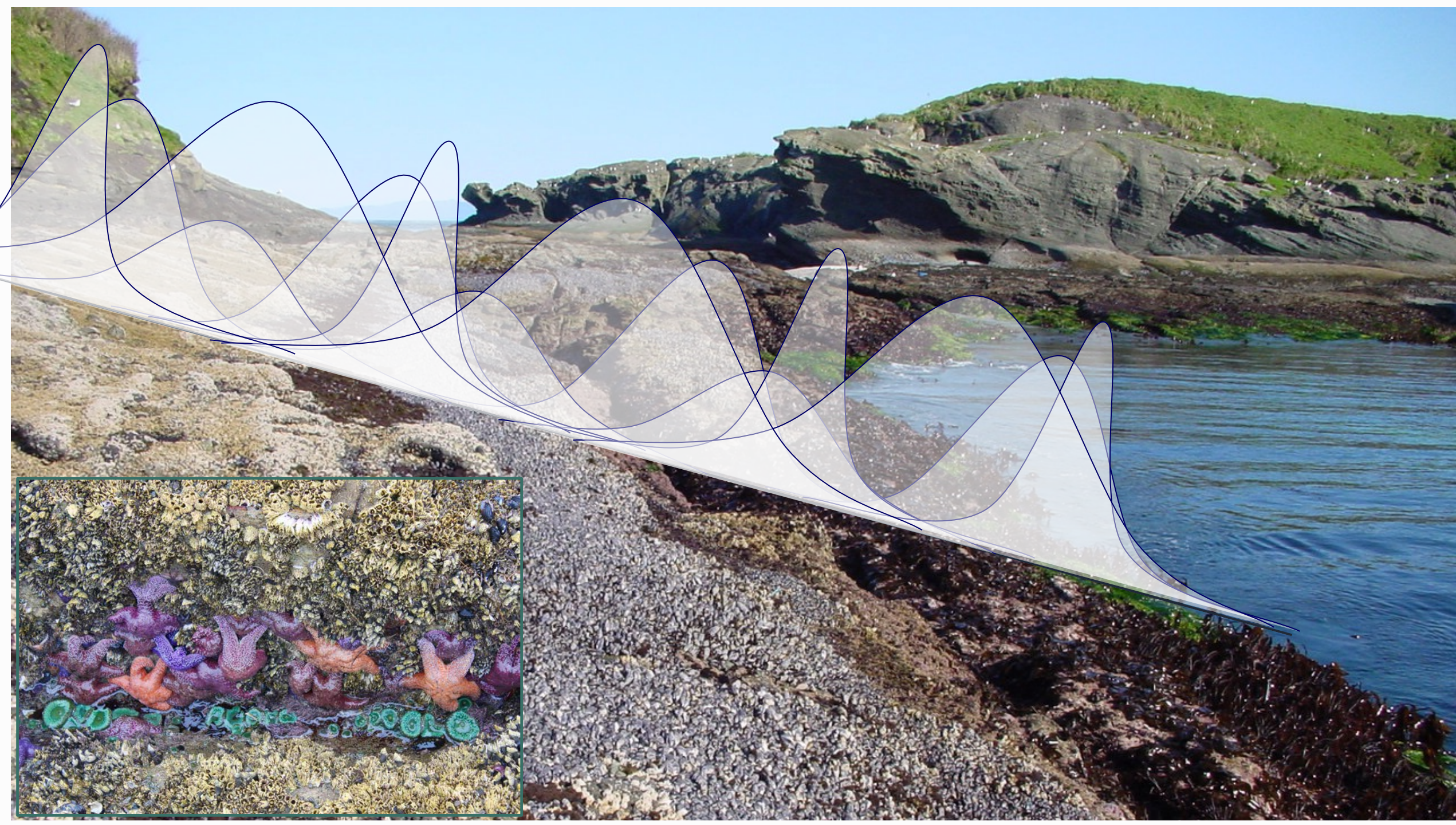


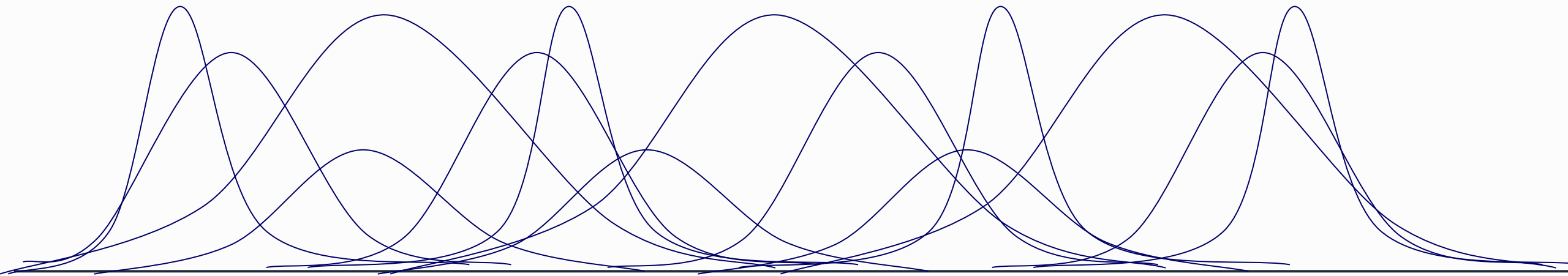
(from Ellenberg 2001)



Environmental gradient







Environmental gradient

Sites

Community data

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achimill		4	4			1																			1					3				
Agrocani							4	5									6			1														
Agrocapi	3	4	2													8				1	1						4		5	3				
Agrorepe		1	1		7	5			4		4					9	3			3	5					5			9	5				
Agrostol					6				4		12									5		1				2	3					8		
Alopgeni									4	1										2				3		5					6			
Alopprat		1			7	7	1		2		5		1	1	4	2			4	5		3	2		5	1	1	3	2					
Anthodor	6					5							1				1				1						4			4				
Caltpalu				1													1	5					6											
Cardprat				2			2	1					1	3					1	1	2	1							1					
Caredist																		1			2													
Caregrac				4			5	8		1			7	2			4	3				1	2			2							1	
Careprae		2	2		1	1										2			1						2		1	3	1					
Carevesi			4			1	4										1	7				2												
Carevulp							1		1		1		1	1					3		1				1									
Cirsarve							1							4						3											1			
Cirslanc																							1											
Cniddubi																													2					
Desccesp															8		1		5	4									3	1				
Eleounig											1	2							1		1													
Eropvern	1					1																					1	1						
Euphesul			1																									1		1				
Festprat							2	1																					1					
Galipalu				1			2	1		1	1	3	2					3		1		1	3	1	1	2						1		
GaliverA	1	1	3											4													4		1	4				
Glycflui							1												2						7						2			
Glycmaxi							1				2												1	2	3		1			1	11	10		
Holclana							4	2						4			1			2	6									1				
Junceffu			1			1	1							1							4	1				1								
Lathprat												1	1								1									1				
Lotucorn		1	1		1																	1								1				
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Phalarun			5			1	2	2	1	11	8	2					1		1	2	1	1	6			1						1		
Planinte									3	3	1									1						1								
Poa palu				1		1	1		11		4		1		1		1		6	2		4	1		6	3								
Poa praA	3	2		1	5	6	1							3		1			1		4		2		2			4	2					
Poa triv			1		2	1	1	4												2		2	3		3	1								
Poteanse			1			2									1			1			2													
Poterept								1											1							1								
Ranufiam				1			1	3					1									1					3							
Ranurepe			4	1	1	1	2		4	1	4		1	3			3		5	4	4	2	6		9	9								
Roriamph									4		7																							
RorisylA							1	2	1							1			1				3		4									
Rumeacet				1			4								4								4					1			1			
Rumethyr	1	5	2			3	5										3									2		4			1			
Siumlati				1			2				1							3					2	1			1							
Stelpalu				2			1				1		1	1			1	1	2			1	1											
Sympoffi																							3	1										
TaraoffA			1	1	1	1					1					1			1	1		1			2				1	1				
Trifrepe				2			3												4	4	1	1												
Vicicrac				1			1				9		4																	1	2			
Vicilath		1																																
Vicitetr		1	1										2														1	1						

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
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Agrocapri	3	4	2													8					1	1						4		5	3			
Agrorepe		1	1		7	5				4		4				9	3			3	5					5			9	5				
Agrostol						6				4		12								5			1				2	3					8	
Alopgeni										4	1									2					3		5					6		
Alopprat			1		7	7	1			2		5		1	1	4	2			4	5		3	2		5	1	1	3	2				
Anthodor	6						5							1			1					1						4			4			
Caltpalu				1													1	5					6											
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Caredist																		1			2													
Caregrac				4			5	8		1			7	2			4	3					1	2			2						1	
Careprae	2	2			1	1										2			1							2		1	3	1				
Carevesi				4			1	4									1	7					2											
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Cirslanc																							1											
Cniddubi																													2					
Desccesp													8				1		5			4								3	1			
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Eropvern	1					1																						1	1					
Euphesul			1																										1		1			
Festprat							2	1																						1				
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Holclana							4	2						4			1			2	6		1	2	3		1							
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Poa palu				1		1	1		11		4		1		1		1		6	2		4	1		6	3								
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Poa triv				1		2	1	1	4											2			2	3		3	1							
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Poterept									1									1								1								
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Rumeacet				1			4							4								4										1		
Rumethyr	1	5	2		3	5										3										2		4			1			
Siumlati				1			2					1					3						2	1			1							
Stelpalu				2			1				1		1	1			1	1	2			1	1											
Sympoffi																							3	1										
TaraoffA				1	1	1	1					1				1			1	1		1			2				1	1				
Trifrepe				2			3												4	4	1	1												
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Vicitetr		1	1											2														1	1					



Species

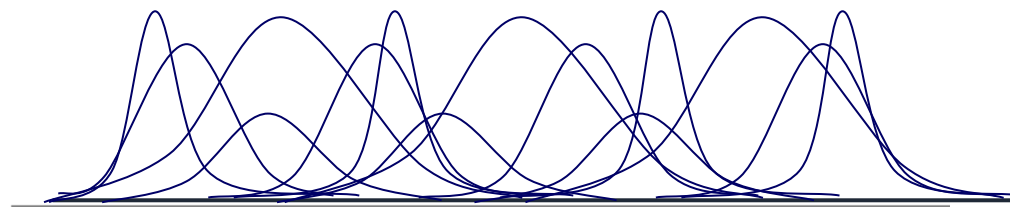
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GaliverA	1	1	4	3		4	1					4																							
Eropvern	1		1					1			1																								
Rumethyr	1	5	4	2	3	1			3		5				2																				
Vicitetr		1	1	1				1				2																							
Cnididubi						2																													
Careprae		2	1	2	2	1	3	1			1				2			1																	
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Rumeacet			1			1						4	4			4			1																
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Poteanse												1		2		2			1					1											
Lychflos												1	1			2			1						1										
Trifrepe													1	4		3		4	2	1															
Caredist												2												1											
Junceffu												1	4			1			1	1							1	1							
Carevulp												1	1		1		1	3				1	1												
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Stelpalu												1	1			1	1	2	2	1	1		1	1											
Poa triv											2			2	3	1			1	2		4				3	1	1							
Poa palu											1	1		2	6	1	4	6	1	4	1	11	1		1		3								
Eleounig													1				1	1					2												
Ranurepe									1		1	3	4	4	9	1	4	5	4	2	3	4	1			6	2	9	1						
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Agrocani												1				4					6						5								
Cirslanc																					1														
Sympoffi																					3				1										
Caltpalu																				1	6	1													
Planinte																	1	1	1				3												
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Caregrac												2					5			4	1	4		7	3	2	8	2	1				1		
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Agrostol												6					2			5		1		4				3	12	8					
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Aloppeni																			2				4					5	1	6		3			
Roriamp																																			
Glycflui																																			
Glycmaxi																																			

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achimill		4	4			1																				1					3			
Agrocani							4	5								6				1														
Agrocapi	3	4	2													8				1	1							4		5	3			
Agrorepe		1	1		7	5			4		4				9	3			3	5					5				9	5				
Agrostol						6			4		12								5			1				2	3					8		
Alopgeni									4	1									2					3		5						6		
Alopprat		1			7	7	1		2		5		1	1	4	2			4	5		3	2		5	1	1	3	2					
Anthodor	6						5						1				1				1						4			4				
Caltpalu				1													1	5					6											
Cardprat			2				2	1					1	3					1	1	2	1							1					
Caredist																		1			2													
Caregrac				4			5	8		1			7	2		4	3					1	2			2							1	
Careprae	2	2			1	1										2			1						2		1	3	1					
Carevesi			4				1	4									1	7				2												
Carevulp									1		1		1	1				3		1					1									
Cirsarve							1							4						3										1				
Cirslanc																							1											
Cniddubi																													2					
Desccesp													8				1		5		4								3	1				
Eleounig											1	2							1		1													
Eropvern	1					1																					1	1						
Euphesul			1																									1		1				
Festprat							2	1																						1				
Galipalu				1			2	1		1	1	3	2					3		1		1	3	1	1	2						1		
GaliverA	1	1	3										4														4		1	4				
Glycflui									1										2													2		
Glycmaxi									1			2																			1	11	10	
Holclana							4	2					4				1		2	6			1	2	3		1							
Junceffu			1				1	1					1							4	1					1						1		
Lathprat													1	1							1									1				
Lotucorn		1	1		1																	1								1				
Lychflos				1			2							1							1													
Phalarun				5			1	2	2	1	11	8	2				1		1	2	1	1	6			1						1		
Planinte									3	3	1								1							1								
Poa palu				1		1	1		11		4		1		1		1		6	2		4	1		6	3								
Poa praA	3	2		1	5	6	1							3		1			1		4		2		2			4	2					
Poa triv				1		2	1	1	4											2		2	3		3	1								
Poteanse				1		2								1				1		2														
Poterept									1									1								1								
Ranufiam				1			1	3						1								1					3							
Ranurepe				4	1	1	1	2	4	1	4		1	3			3		5	4	4	2	6		9	9								
Roriamph									4		7																							
RorisylA								1	2	1					1								3		4		1							
Rumeacet				1			4							4								4									1			
Rumethyr	1	5	2		3	5										3									2		4				1			
Siumlati				1				2				1					3					2	1			1								
Stelpalu				2			1				1	1	1				1	1	2			1	1											
Sympoffi																							3	1										
TaraoffA				1	1	1	1				1					1			1	1		1			2				1	1				
Trifrepe				2			3												4	4	1	1												
Vicicrac				1			1				9			4																1	2			
Vicilath		1																																
Vicitetr		1	1											2													1	1						



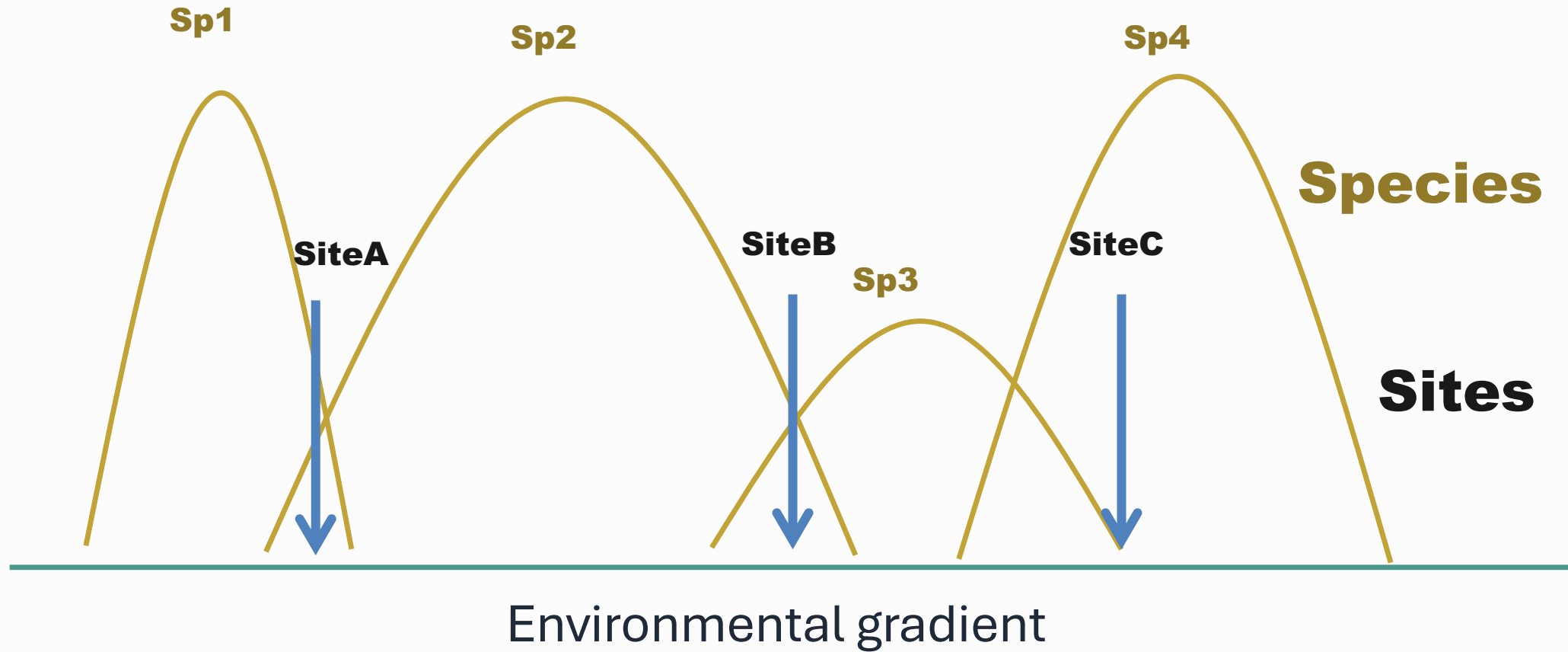
Species

[illegible]



Ordination - Correspondence Analysis

What is it trying to do?



If you knew the species or the sites you could position the other

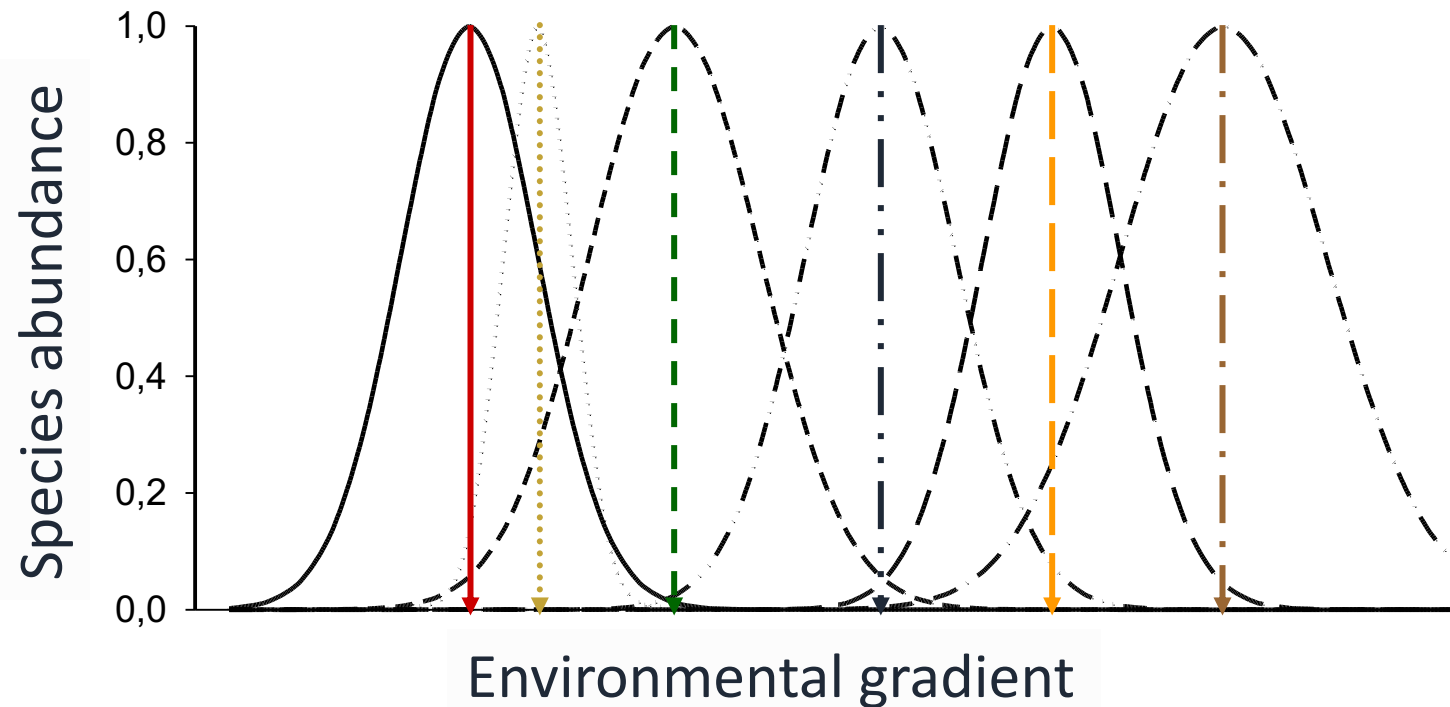
Correspondence Analysis



Assumes a **unimodal** relationship of species along important gradients

The principle:

Estimates the **optimum** of a species using a **weighted average**



[illegible]

Correspondence Analysis



	1	2	3	16	30	29	28	5	15	6	14	21	20	25	7	11	19	4	22	17	9	13	18	23	8	26	10	31	12	24	33	32	GM			
Vicilath	1																																-1.441			
Achimill		4	4		3					1			1																					-1.218		
Agrocapi	3	4	4	2	8	3	5					1								1														-1.217		
Euphesul			1		1		1																											-1.194		
GaliverA	1	1	4	3		4	1				4																							-1.141		
Eropvern	1	1						1		1																								-1.117		
Rumethyr	1	5	4	2	3	1			3	5				2																				-1.083		
Vicitetr		1	1	1				1			2																								-1.027	
Cniddubi							2																											-0.967		
Careprae		2	1	2	2		1	3	1		1			2			1																	-0.955		
Anthodor	6		4								1	1			5						1													-0.946		
Lotucorn				1	1						1	1							1															-0.653		
Poa praA		3	2	1		2	4	5		6	3	4		2	1		1	1							2									-0.633		
Agrorepe		1	1	3		5	9	7	9	5			5	5		4	3					4												-0.574		
TaraoffA			1	1	1	1	1	1	1				1	2		1	1	1	1															-0.492		
Cirsarve					1					4	3	1																							-0.440	
Rumeacet			1		1					4	4		4				1																		-0.400	
Lathprat				1						1	1											1													-0.391	
Alopprat		1	1		2		2	3	7	4	7	1		5	5	1	5	4		3		2	1		2		1							-0.378		
Descocesp					1	3					8	5								4	1														-0.357	
Vicicrac					2	1				4				1	9		1																		-0.287	
Holclana				1							4	6	2		4						1					2									-0.219	
Festprat					1										2											1									-0.186	
Cardprat					1						3	2	1		2		1	2	1				1		1										-0.117	
Poteanse											1		2		2				1																-0.091	
Lychflos											1	1			2								1												-0.059	
Trifrepe												1	4		3		4	2	1																-0.034	
Caredist											2												1												-0.015	
Junceffu											1	4			1				1	1						1	1								-0.001	
Carevulp											1	1			1	3					1	1													0.021	
Poterept													1			1					1														0.072	
Stelpalu											1	1			1	1	2	2	1	1		1	1												0.097	
Poa triv											2			2	3	1			1	2		4			3	1	1								0.099	
Poa palu											1	1		2	6	1	4	6	1	4	1	1	1	1	1		3								0.117	
Eleounig													1			1	1						2												0.124	
Ranurepe											1	1	3	4	4	9	1	4	5	4	2	3	4	1		6	2	9	1						0.131	
RorisylA											1				4		1	1				1			3			2							0.164	
Agrocani												1			4						6					5									0.217	
Cirsilanc																					1														0.225	
Sympoffi																						3			1										0.295	
Caltpalu																						1	6	1		5									0.322	
Planinte																1	1						3					3							0.361	
Carevesi																	1		4	2	1			7		4									0.363	
Ranufflam												1			1		1					1				3	3								0.383	
Caregrac													2		5			4	1	4			7	3	2	8	2	1				1			0.399	
Phalarun														1	2		1	1	5	1	1	2	2		6	2	1	1	1	8				0.453		
Siumlati																																				0.476
Agrostol																																				0.512
Galipalu																																				0.577
Aloppeni																																				1.009
RoriampH																																				1.298
Glycflui																																				1.753
Glycmaxi																																				3.117

The principle:

Calculate the **weighted average of species** from **site** values (current order of sites)

$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

x_j = Weighted average of **species j**

n = Number of **sites**

y_{ij} = Abundance of species j in **site i**

x_i = Site score of **site i**



CA – How it Works

Raw data

samples	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
<i>Achmil</i>	1	3			2	2	2			4							2			
<i>Agrsto</i>			4	8				4	3			4	5	4	4	7				5
<i>Airpra</i>																		2	3	
<i>Alogen</i>		2	7	2				5	3			8	5				4			
<i>Antodo</i>					4	3	2			4								4	4	
<i>Belper</i>		3	2	2	2					2									2	
<i>Brohor</i>		4		3	2		2			4										
<i>Chealb</i>													1							
<i>Cirarv</i>				2																
<i>Elepal</i>								4						4	5	8				4
<i>Elyrep</i>	4	4	4	4	4				6											
<i>Empnig</i>																			2	
<i>Hyprad</i>											2						2		5	
<i>Junart</i>								4	4						3	3				4
<i>Junbuf</i>							2		4			4	3							
<i>Leoaut</i>		5	2	2	3	3	3	3	2	3	5	2	2	2	2		2	5	6	2
<i>Lolper</i>	7	5	6	5	2	6	6	4	2	6	7								2	
<i>Plalan</i>					5	5	5			3	3						2	3		
<i>Poapra</i>	4	4	5	4	2	3	4	4	4	4	4		2				1	3		
<i>Poatri</i>	2	7	6	5	6	4	5	4	5	4		4	9			2				
<i>Potpal</i>														2	2					
<i>Ranfla</i>								2					2	2	2	2				4
<i>Rumace</i>					5	6	3		2			2								
<i>Sagpro</i>				5				2	2		2	4	2						3	
<i>Salrep</i>																		3	3	5
<i>Tripra</i>					2	5	2													
<i>Trirep</i>		5	2	1	2	5	2	2	3	6	3	3	2	6	1			2	2	
<i>Violat</i>										1	2							1		
<i>Brarut</i>			2	2	2	6	2	2	2	2	4	4			4	4		6	3	4
<i>Calcus</i>														4		3				3
<i>Hiprha</i>									1									2	1	
<i>Poaann</i>	3	3		4			2		2		2							4		
<i>Ranacr</i>						3	2		2	3			3	1	1					



CA – How it Works

Calculating weighted mean of **species**

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20
Achmil	1	3	2	2	2				4	8	4	2								
Agrsto								4	8				3	4	4	5	4	4	7	5
Airpra											2								3	
Alogen		2						7	2				3	8	5	5		4		
Antodo			4	3	2					4	4								4	
Belper		3	2				2	2	2	2										
Brohor		4	2		2				3	4										
Chealb															1					
Cirarv									2											
Elepal														4		4	5	8		4
Elyrep	4	4	4					4	4			6								
Empnig																			2	
Hyprad						2					2								5	
Junart												4		4			3	3		4
Junbuf															3					
Leoaut															2	2	2		6	2
Lolper	7																			
Plalan																				
Poapra	4																			
Poatri	2																			
Potpal																				
Ranfla																				
Rumace			5	6	3							2	2							4
Sagpro						2			5			2	4	2	2				3	
Salrep							3												3	5
Tripra			2	5	2															
Trirep		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2	
Viclat						2	1			1										
Brarut			2	6	2	4	6	2	2	2		2	4	2			4	4	3	4
Calcus																4		3		3
Hiprha							2					1							1	
Poaann	3	3			2	2	4		4			2								
Ranacr				3	2					3		2			3	1	1			

1. Iteration
Site score is random =
e.g. order of plots

$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

5,9
27
4,6
16
12
4,3
4,2
0,7
0,4
19
5,7
1,9
7,6
12
6,9
30
18
12
17
23
2,9
11
6,2
10
11
2,7
22
2,5
29
8,2
3,2
7,6
7,4



CA – How it Works

Sort species by weighted species scores

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20	
Cirarv								2													0,4
Chealb															1						0,7
Empnig																			2		1,9
Viclat						2	1			1											2,5
Tripra			2	5	2																2,7
Potpal																	2	2			2,9
Hiprha							2					1							1		3,2
Brohor		4	2		2				3	4											4,2
Belper		3	2				2	2	2	2											4,3
Airpra											2								3		4,6
Elyrep	4	4	4					4	4			6									5,7
Achmil	1	3	2	2	2					4	2										5,9
Rumace			5	6	3							2	2								6,2
Junbuf					2							4	4								6,9
Ranacr				3	2					3		2			3	1	1				7,4
Hyprad						2					2								5		7,6
Poaann	3	3			2	2	4		4			2									7,6
Calcus																4		3	3		8,2
Sagpro						2			5			2	4	2	2				3		10
Salrep							3												3	5	11
Ranfla														2	2	2	2	2	4		11
Antodo			4	3	2					4	4								4		12
Junart												4		4			3	3	4		12
Plalan			5	5	5	3	3			3	2										12
Alogen		2						7	2			3	8	5	5			4			16
Poapra	4	4	2	3	4	4	3	5	4	4	1	4		4	2						17
Lolper	7	5	2	6	6	7	2	6	5	6		2		4							18
Elepal														4		4	5	8	4		19
Trirep		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2		22
Poatri	2	7	6	4	5			6	5	4		5	4	4	9			2			23
Agrsto								4	8			3	4	4	5	4	4	7	5		27
Braruf			2	6	2	4	6	2	2	2		2	4	2			4	4	3	4	29
Leoaut		5	3	3	3	5	5	2	2	3	2	2	2	3	2	2	2		6	2	30



CA – How it Works

Calculating weighted mean of sites

samples	1	2	5	6	7	11	18	3	4	10	17	9	12	8	13	14	15	16	19	20	
Cirarv								2													0,4
Chealb															1						0,7
Empnig																			2		1,9
Viclat						2	1			1											2,5
Tripra			2	5	2																2,7
Potpal																	2	2			2,9
Hiprha							2					1							1		3,2
Brohor		4	2		2				3	4											4,2
Belper		3	2				2	2	2	2											4,3
Airpra												2							3		4,6
Elyrep	4	4	4					4	4			6									5,7
Achmil	1	3	2	2	2					4	2										5,9
Rumace			5	6	3							2	2								6,2
Junbuf					2							4	4			3					6,9
Ranacr				3	2					3		2			3	1	1				7,4
Hyprad						2					2								5		7,6
Poaann	3	3			2	2	4		4			2									7,6
Calcus																4		3	3		8,2
Sagpro						2			5			2	4	2	2				3	5	10
Salrep							3												3	5	11
Ranfla														2	2	2	2	2	4	4	11
Antodo			4	3	2					4	4								4		12
Junart												4		4			3	3	4	4	12
Plalan			5	5	5	3	3			3	2										12
Alogen		2						7	2			3	8	5	5			4			16
Poapra	4	4	2	3	4	4	3	5	4	4	1	4		4	2						17
Lolper	7	5	2	6	6	7	2	6	5	6		2		4							18
Elepal														4		4	5	8		4	19
Trirep		5	2	5	2	3	2	2	1	6		3	3	2	2	6	1		2		22
Poatri	2	7	6	4	5			6	5	4		5	4	4	9			2			23
Agrsto								4	8			3	4	4	5	4	4	7		5	27
Brarut			2	6	2	4	6	2	2	2		2	4	2			4	4	3	4	29
Leoaut		5	3	3	3	5	5	2	2	3	2	2	2	3	2	2	2		6	2	30
	1,6	2,8	2,7	2,6	2,7	1,7	2,4	1,0	3,8	2,2	1,1	3,9	2,5	0,6	1,9	1,4	0,4	0,7	3,7	2,3	

$$\tilde{x}_i = \frac{\sum_{j=1}^n \tilde{x}_j y_{ij}}{\sum_{j=1}^n y_{ij}}$$



CA – How it Works

Sort by weighted site scores

samples	15	8	16	3	17	14	1	11	13	10	20	18	12	6	7	5	2	19	4	9	
Cirarv																			2		0,2
Chealb									1												0,3
Empnig																		2			0,3
Viclat								2		1		1									0,5
Tripra														5	2	2					0,5
Potpal	2					2															0,6
Hiprha												2						1		1	1
Brohor										4					2	2	4		3		1
Belper				2						2		2				2	3		2		1
Airpra					2													3			1,1
Elyrep				4			4									4	4		4	6	1,1
Achmil					2		1			4					2	2	2	3			1,5
Rumace														2	6	3	5			2	1,5
Junbuf									3					4		2				4	1,6
Ranacr	1					1			3	3					3	2				2	1,6
Hyprad					2			2										5			2,2
Poaann							3	2				4				2		3		4	2,2
Calcus			3			4					3										2,3
Sagpro		2						2	2				4					3	5	2	2,5
Salrep											5	3						3			2,5
Ranfla	2	2	2			2			2		4										2,6
Antodo					4					4					3	2	4		4		3,5
Junart	3	4	3								4									4	3,6
Plalan					2			3		3		3		5	5	5					4,3
Alogen		5	4	7					5				8				2		2	3	4,9
Poapra		4		5	1		4	4	2	4		3		3	4	2	4		4	4	4,9
Lolper		4		6			7	7		6		2		6	6	2	5		5	2	6,3
Elepal	5	4	8			4					4										6,3
Trirep	1	2		2		6		3	2	6		2	3	5	2	2	5	2	1	3	6,7
Poatri		4	2	6			2		9	4			4	4	5	6	7		5	5	7,1
Agrsto	4	4	7	4		4			5		5		4						8	3	7,2
Braruf	4	2	4	2				4		2	4	6	4	6	2	2		3	2	2	8,3
Leoaut	2	3		2	2	2		5	2	3	2	5	2	3	3	3	5	6	2	2	9,9
	0,4	0,6	0,7	1,0	1,1	1,4	1,6	1,7	1,9	2,2	2,3	2,4	2,5	2,6	2,7	2,7	2,8	3,7	3,8	3,9	



CA – How it Works

Further iterations until no changes

samples	17	15	14	20	19	16	1	18	12	11	13	5	8	7	9	10	6	4	2	3	
Emprig					2																0,5
Chealb											1										0,6
Potpal		2	2																		1
Airpra	2				3																1,1
Cirarv																		2			1,4
Hiprha					1			2							1						1,7
Viclat								1		2						1					2,1
Hyprad	2				5					2											2,7
Calcus			4	3		3															2,9
Salrep				5	3			3													3,4
Ranfla																					-
Tripra																					
Junbuf																					
Junart																					
Belper																					
Elepal																					
Ranacr																					
Achmil																					
Antodo																					
Brohor																					
Sagpro					3				4	2	2							3	4		10
Rumace									2			5		3	2		6				11
Poaann								3	4		2				2			4	3		11
Plalan	2								3		3		5		5		3	5			15
Elyrep								4				4			6			4	4	4	16
Alogen						4			8		5		5		3			2	2	7	21
Agrsto		4	4	5		7			4		5		4		3		8		4		23
Brarut		4		4	3	4			6	4	4		2	2	2	2	6	2		2	24
Trirep		1	6		2				2	3	3	2	2	2	2	3	6	5	1	5	26
Leoaut	2	2	2	2	6				5	2	5	2	3	3	3	2	3	3	2	5	28
Poapra	1								4	3		4	2	2	4	4	4	3	4	4	29
Lolper									7	2		7		2	4	6	2	6	6	5	36
Poatri						2	2				4		9	6	4	5	5	4	4	5	39
	5,4	11	11	12	12	16	17	19	23	24	25	26	27	29	29	30	32	32	32	32	

Definition of first axis: further axes are calculated on the same principle but for remaining unexplained variance

Species

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
Achimill		4	4			1																				1					3			
Agrocani							4	5								6				1														
Agrocapi	3	4	2													8				1	1							4		5	3			
Agrorepe		1	1		7	5			4		4				9	3			3	5					5				9	5				
Agrostol						6			4		12								5			1				2	3						8	
Alopgeni									4	1									2					3		5						6		
Alopprat		1			7	7	1		2		5		1	1	4	2			4	5		3	2		5	1	1	3	2					
Anthodor	6						5						1				1				1						4			4				
Caltpalu				1													1	5					6											
Cardprat			2				2	1					1	3					1	1	2	1							1					
Caredist																		1			2													
Caregrac				4			5	8		1			7	2		4	3					1	2			2							1	
Careprae	2	2			1	1										2			1						2		1	3	1					
Carevesi			4				1	4									1	7				2												
Carevulp									1		1		1	1					3		1				1									
Cirsarve							1							4						3											1			
Cirslanc																							1											
Cniddubi																													2					
Desccesp													8				1		5			4								3	1			
Eleounig											1	2							1		1													
Eropvern	1					1																					1	1						
Euphesul			1																									1		1				
Festprat							2	1																						1				
Galipalu				1			2	1		1	1	3	2					3		1		1	3	1	1	2						1		
GaliverA	1	1	3										4														4		1	4				
Glycflui									1										2													2		
Glycmaxi									1			2																				1	11	10
Holclana							4	2					4				1		2	6			1	2	3		1							
Junceffu			1				1	1						1						4	1					1								
Lathprat													1	1							1										1			
Lotucorn		1	1		1																	1								1				
Lychflos				1			2							1							1													
Phalarun				5			1	2	2	1	11	8	2				1		1	2	1	1	6			1						1		
Planinte									3	3	1								1							1								
Poa palu				1		1	1		11		4		1		1		1		6	2		4	1		6	3								
Poa praA	3	2		1	5	6	1							3		1			1		4		2		2			4	2					
Poa triv				1		2	1	1	4											2		2	3		3	1								
Poteanse				1		2								1				1		2														
Poterept									1									1								1								
Ranufлам				1			1	3						1								1					3							
Ranurepe				4	1	1	1	2	4	1	4		1	3			3		5	4	4	2	6		9	9								
Roriamph									4		7																							
RorisylA									1	2	1				1								3		4		1							
Rumeacet				1			4							4								4										1		
Rumethyr	1	5	2		3	5										3										2		4			1			
Siumlati				1				2				1					3					2	1			1								
Stelpalu				2			1				1	1	1				1	1	2			1	1											
Sympoffi																							3	1										
TaraoffA				1	1	1	1				1					1			1	1		1			2				1	1				
Trifrepe				2			3												4	4	1	1												
Vicicrac				1			1				9			4																		1	2	
Vicilath		1																																
Vicitetr		1	1											2													1	1						



Species

[illegible]

Correspondence Analysis



	1	2	3	16	30	29	28	5	15	6	14	21	20	25	7	11	19	4	22	17	9	13	18	23	26	8	10	31	12	24	33	32	GM	
Vicilath	1																																-1.441	
Achimill		4	4	3					1			1																					-1.218	
Agrocapi	3	4	4	2	8	3	5					1								1													-1.217	
Euphesul			1		1		1																										-1.194	
GaliverA	1	1	4	3		4	1				4																						-1.141	
Eropvern	1	1						1		1																							-1.117	
Rumethyr	1	5	4	2	3	1			3	5			2																				-1.083	
Vicitetr		1	1	1				1			2																						-1.027	
Cniddubi							2																										-0.967	
Careprae		2	1	2	2		1	3	1		1		2			1																	-0.955	
Anthodor	6		4		4					1	1			5						1													-0.946	
Lotucorn				1	1					1	1					1																	-0.653	
Poa praA		3	2	1		2	4	5		6	3	4		2	1		1	1						2									-0.633	
Agrorepe		1	1	3		5	9	7	9	5			5	5		4	3					4											-0.574	
TaraoffA			1	1	1	1	1	1	1			1	2		1	1	1	1															-0.492	
Cirsarve				1						4	3	1																						-0.440
Rumeacet		1		1						4	4		4		1																			-0.400
Lathprat				1						1	1											1												-0.391
Alopprat	1	1	2		2	3	7	4	7	1		5	5	1	5	4		3		2	1		2		1								-0.378	
Descocesp				1	3					8	5							4	1															-0.357
Vicicrac				2	1					4			1	9		1																		-0.287
Holclana				1						4	6	2	4					1						2										-0.219
Festprat					1							2												1										-0.186
Cardprat				1						3	2	1	2		1	2	1				1		1											-0.117
Poteanse										1	2	2			1							1												-0.091
Lychflos										1	1	2			1								1											-0.059
Trifrepe										1	4	3		4	2	1																		-0.034
Caredist										2												1												-0.015
Junceffu										1	4		1		1	1							1	1										-0.001
Carevulp										1	1		1	3							1	1												0.021
Poterept											1		1								1													0.072
Stelpalu										1	1		1	2	2	1	1				1	1												0.097
Poa triv										2		2	3	1		1	2		4				3	1	1									0.099
Poa palu										1	1		2	6	1	4	6	1	4	1	1	1		1		3								0.117
Eleounig												1		1								2												0.124
Ranurepe										1	1	3	4	4	9	1	4	5	4	2	3	4	1		6	2	9	1						0.131
RorisylA										1			4		1	1					1		3		2									0.164
Agrocani											1		4					6						5										0.217
Cirsilanc																					1													0.225
Sympoffi																							1											0.295
Caltpalu																																		0.322
Planinte																																		0.361
Carevesi																																		0.363
Ranuffam																																		0.383
Caregrac																																		0.399
Phalarun																																		0.453
Siumlati																																		0.476
Agrostol																																		0.512
Galipalu																																		0.577
Aloppeni																																		1.009
Roriamph																																		1.298
Glycflui																																		1.753
Glycmaxi																																		3.117

The principle:
Calculate the **weighted average of species**
from **site** values (current order of sites)

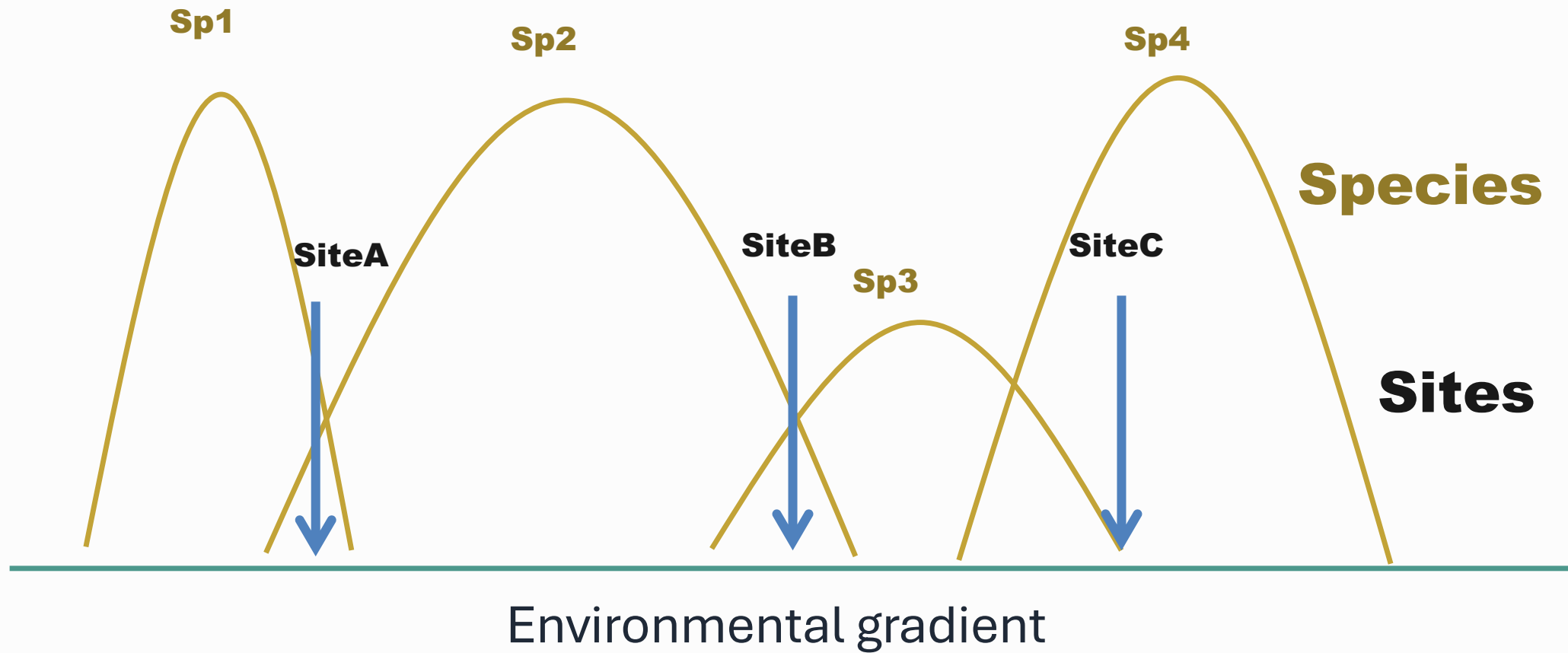
$$\tilde{x}_j = \frac{\sum_{i=1}^n x_i y_{ij}}{\sum_{i=1}^n y_{ij}}$$

- x_j = Weighted average of **species j**
- n = Number of **sites**
- y_{ij} = Abundance of species j in **site i**
- x_i = Site score of **site i**



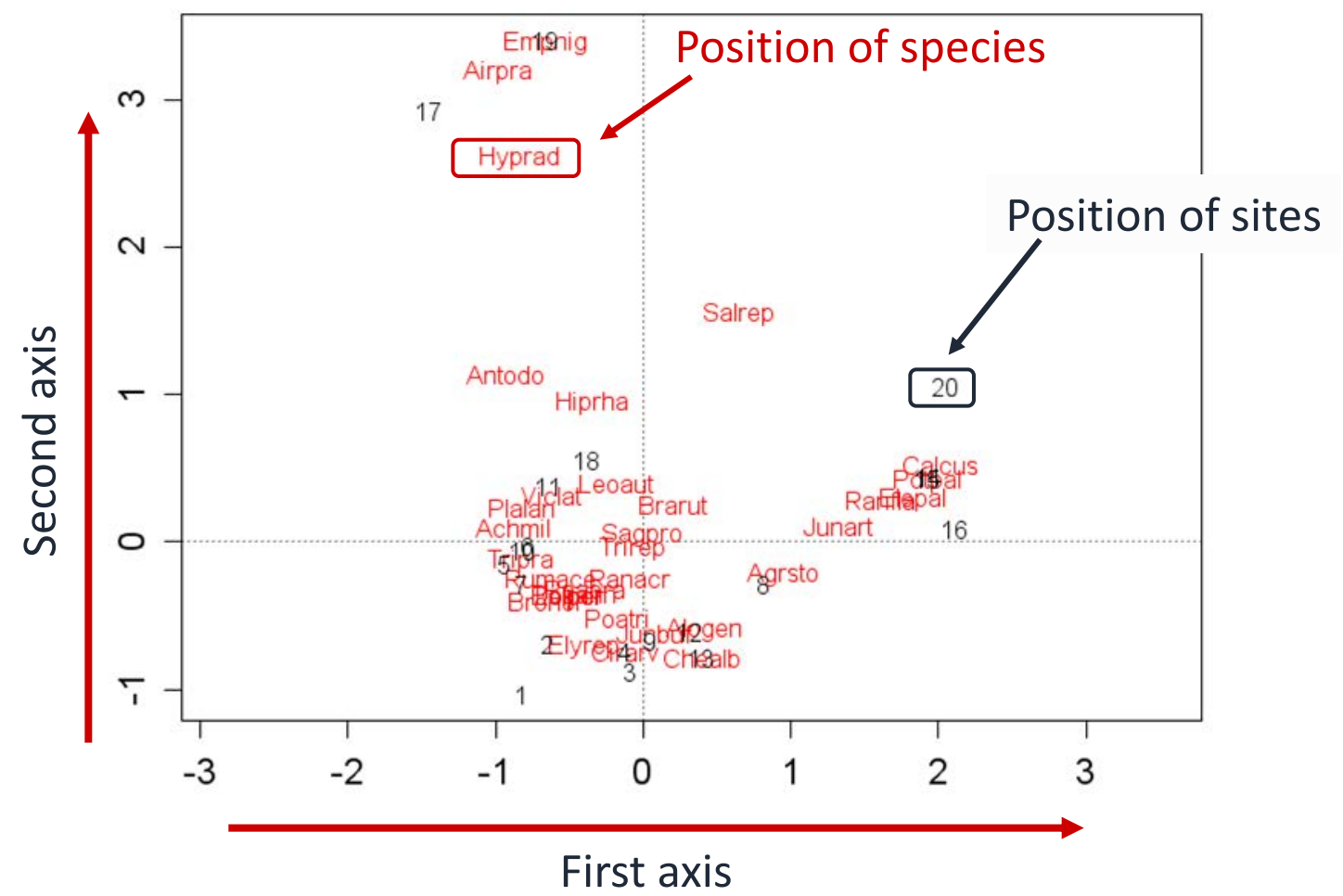
Ordination - Correspondence Analysis

What is it trying to do?

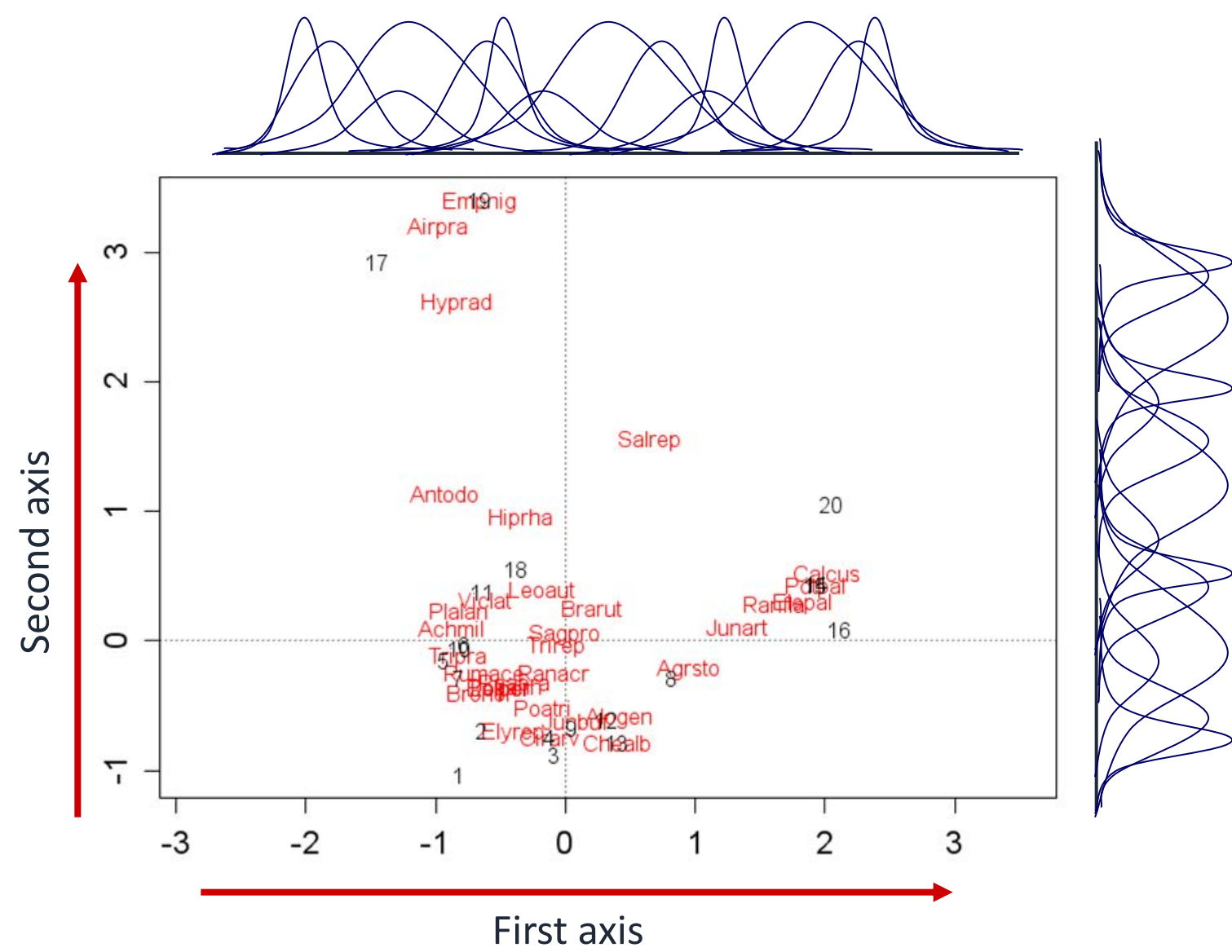


If you knew the species or the sites you could position the other

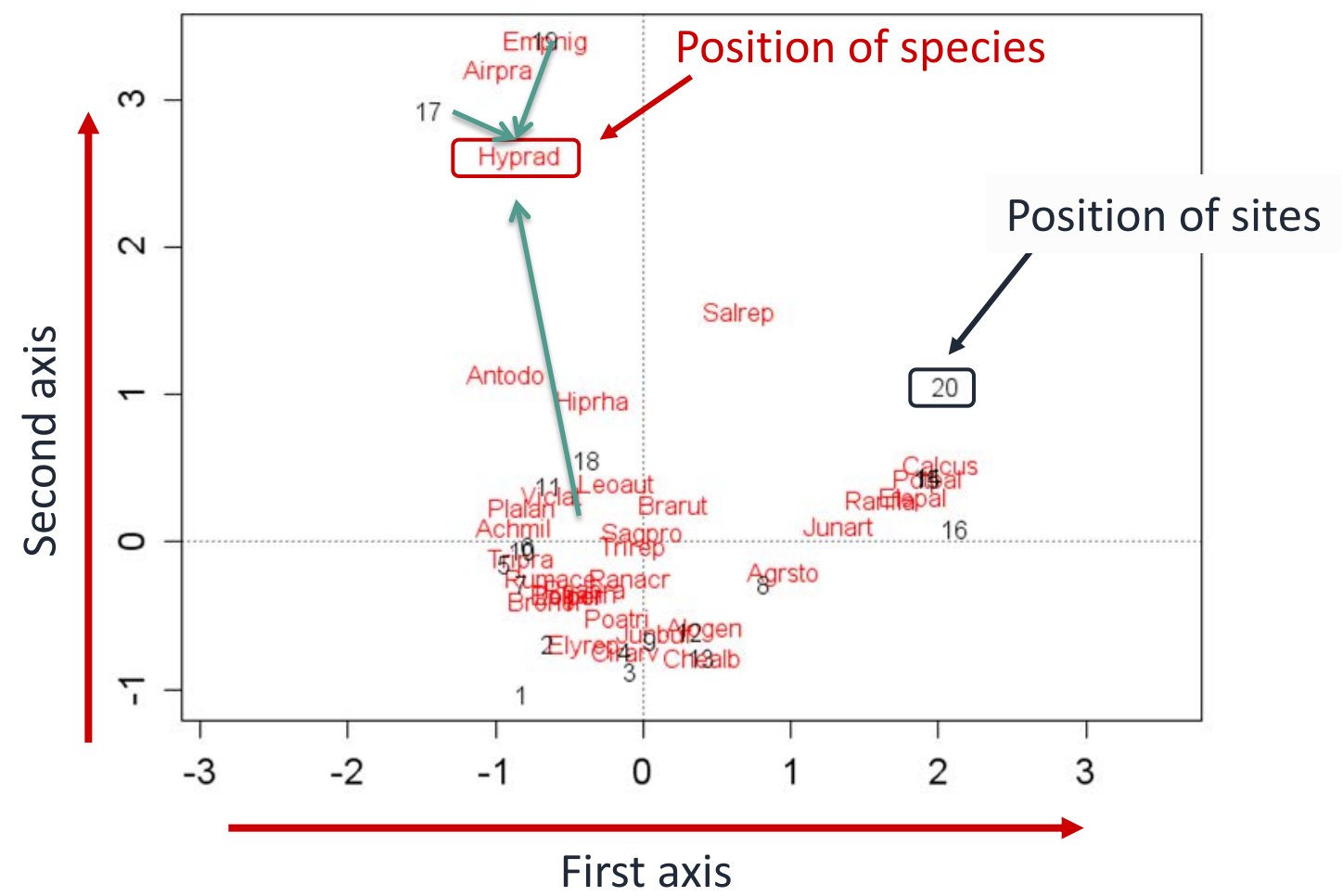
Correspondence Analysis



Correspondence Analysis



Correspondence Analysis





Correspondence Analysis

Each axis of the CA is an **eigenvector**

Each axis has an **eigenvalue**

- Measures the degree of differentiation between species along this axis (range 0-1)
- The larger, the more species can be differentiated along this axis

	Inertia	
Total	2.156	← total inertia: Overall variance of the data
Unconstrained	2.156 19	

Eigenvalues for unconstrained axes:

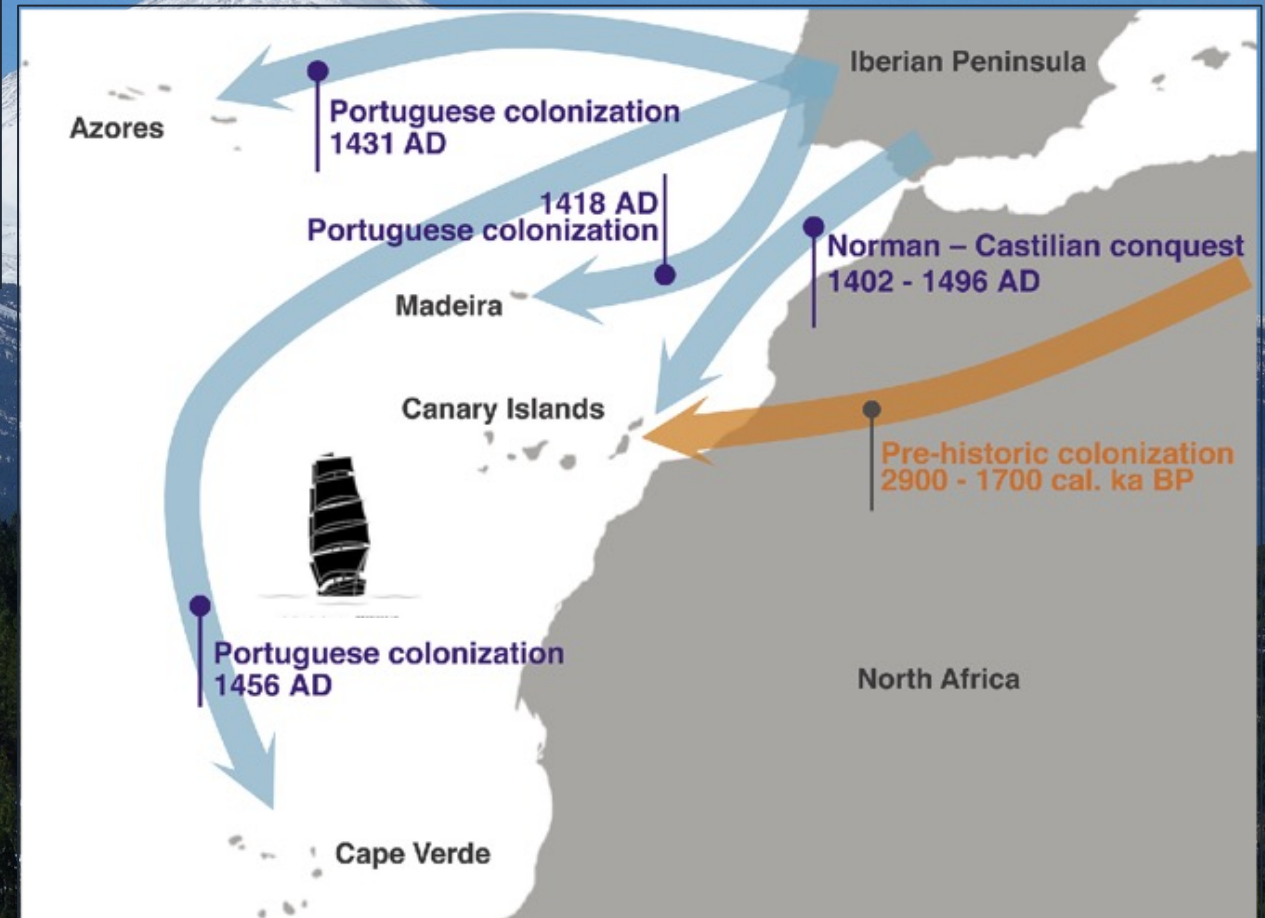
CA1	CA2	CA3	CA4	CA5	CA6
0.52722	0.39537	0.25162	0.20418	0.14553	0.10627

Decline in explained variance with increasing ordination axes

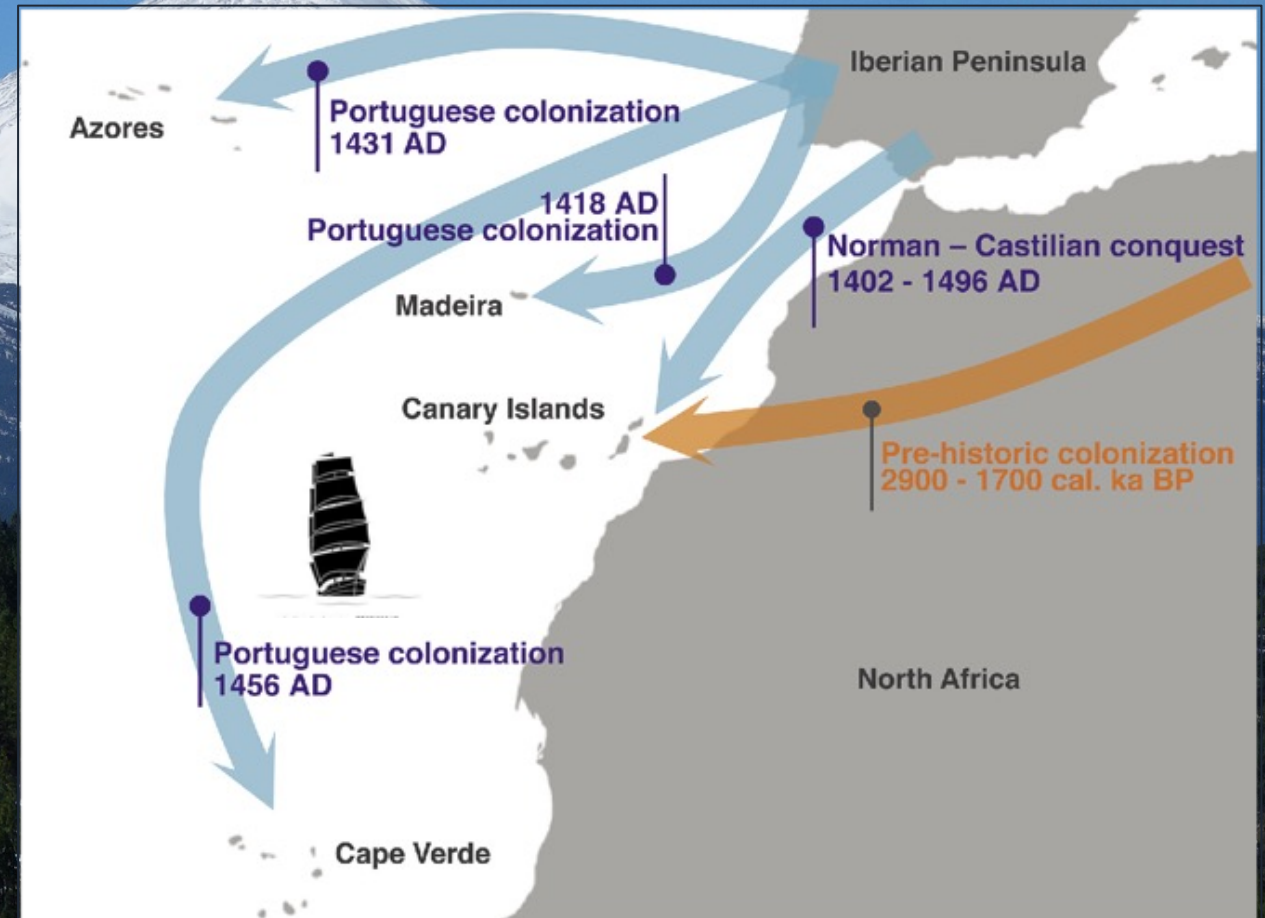


Tenerife, 3718m, Canary Islands, from Christoph Hormann, imagico.de

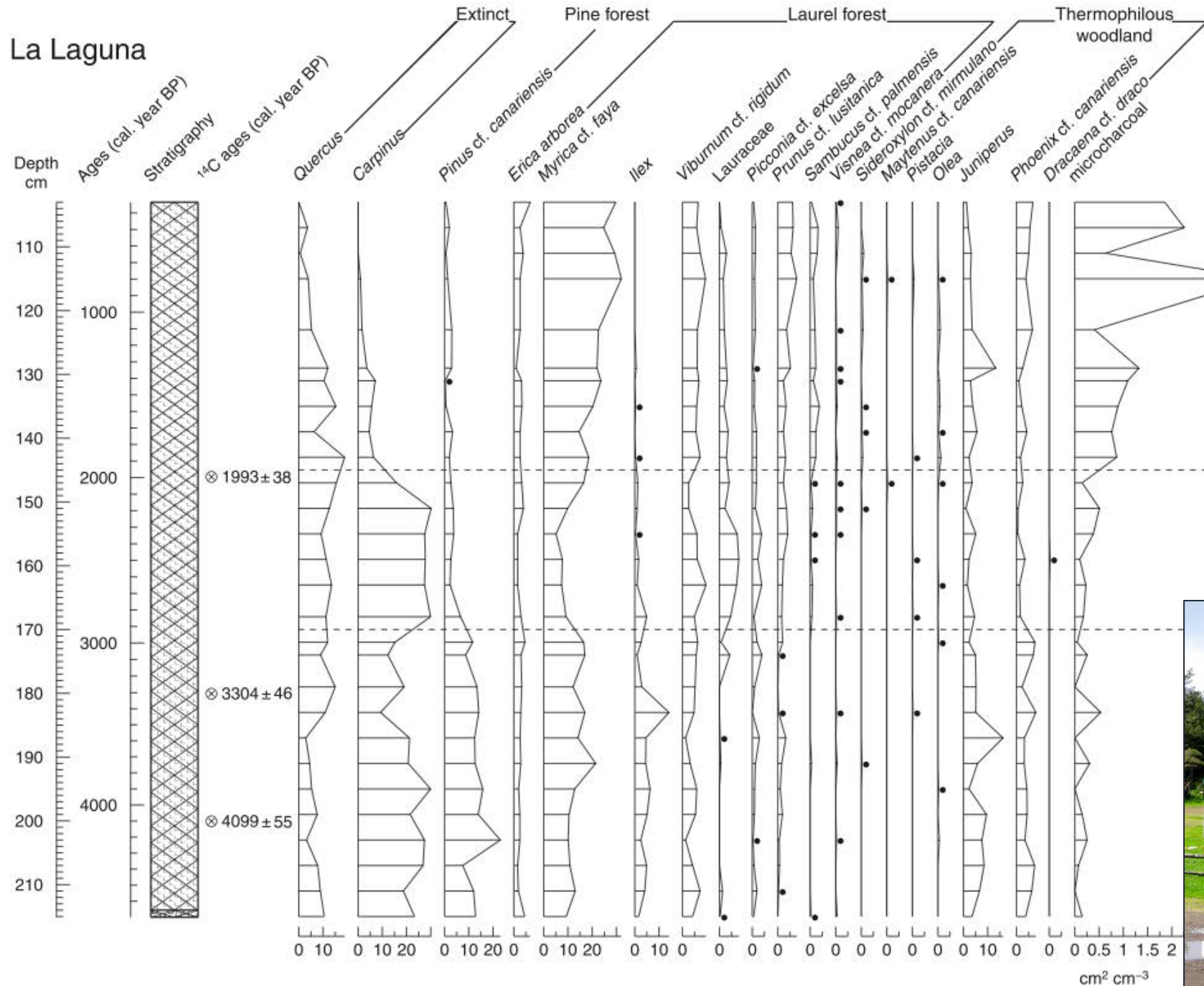
Islands



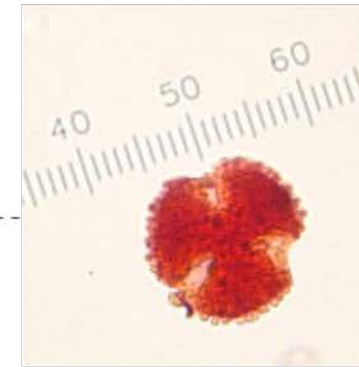
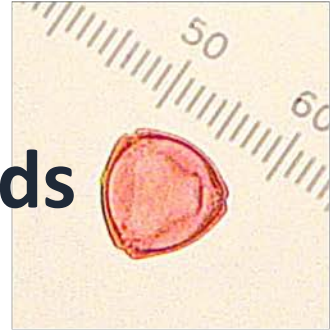
Islands



La Laguna

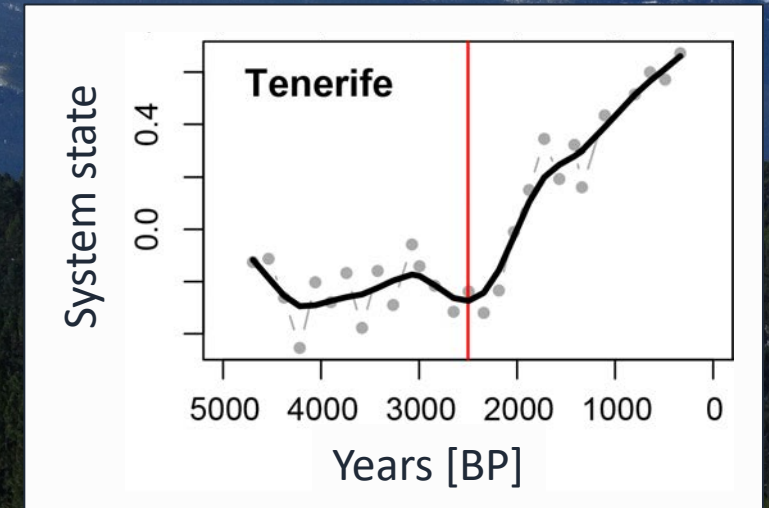


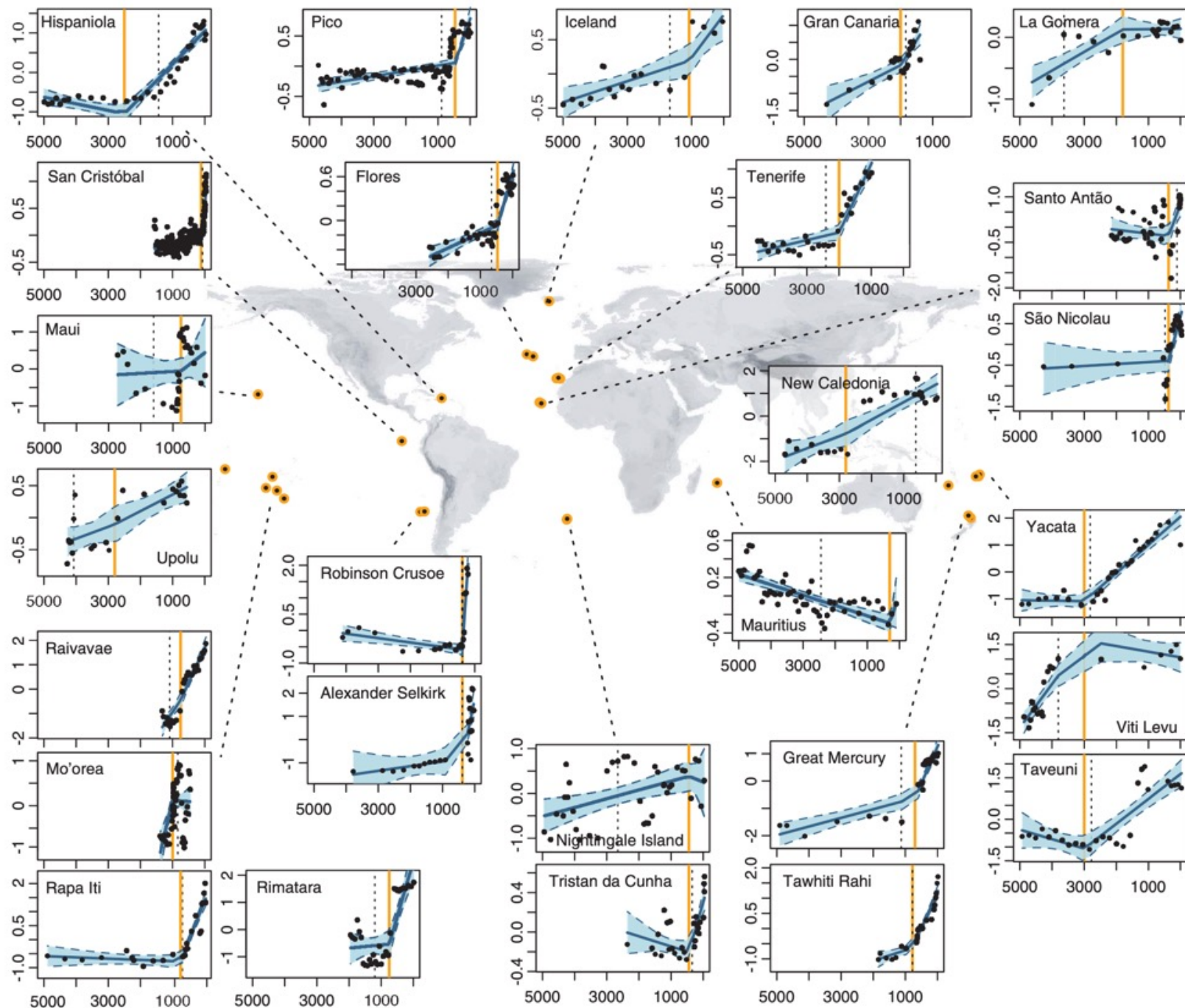
Tenerife, Canary Islands





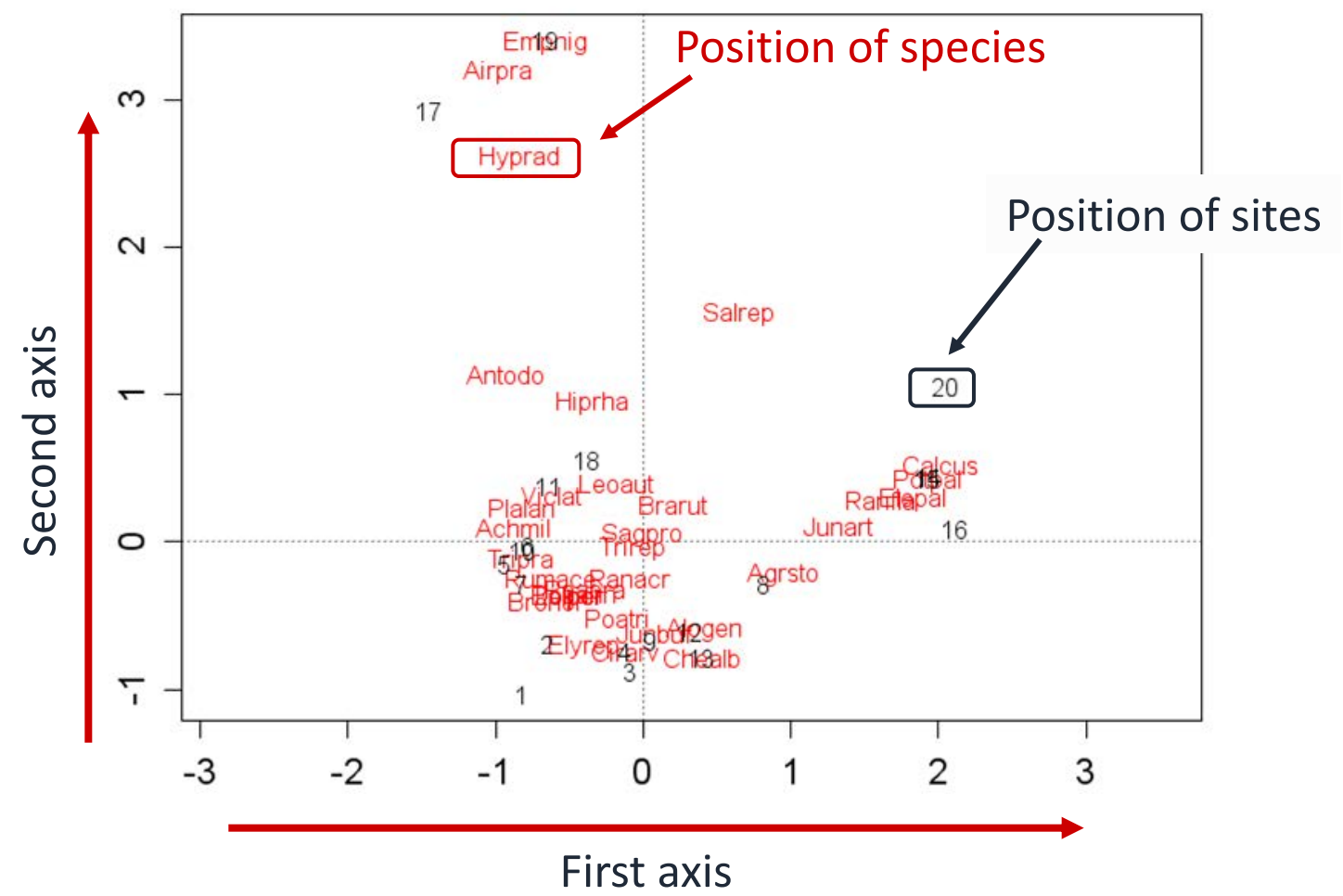
Islands





Nogue et al. (Science, 2021)

Correspondence Analysis

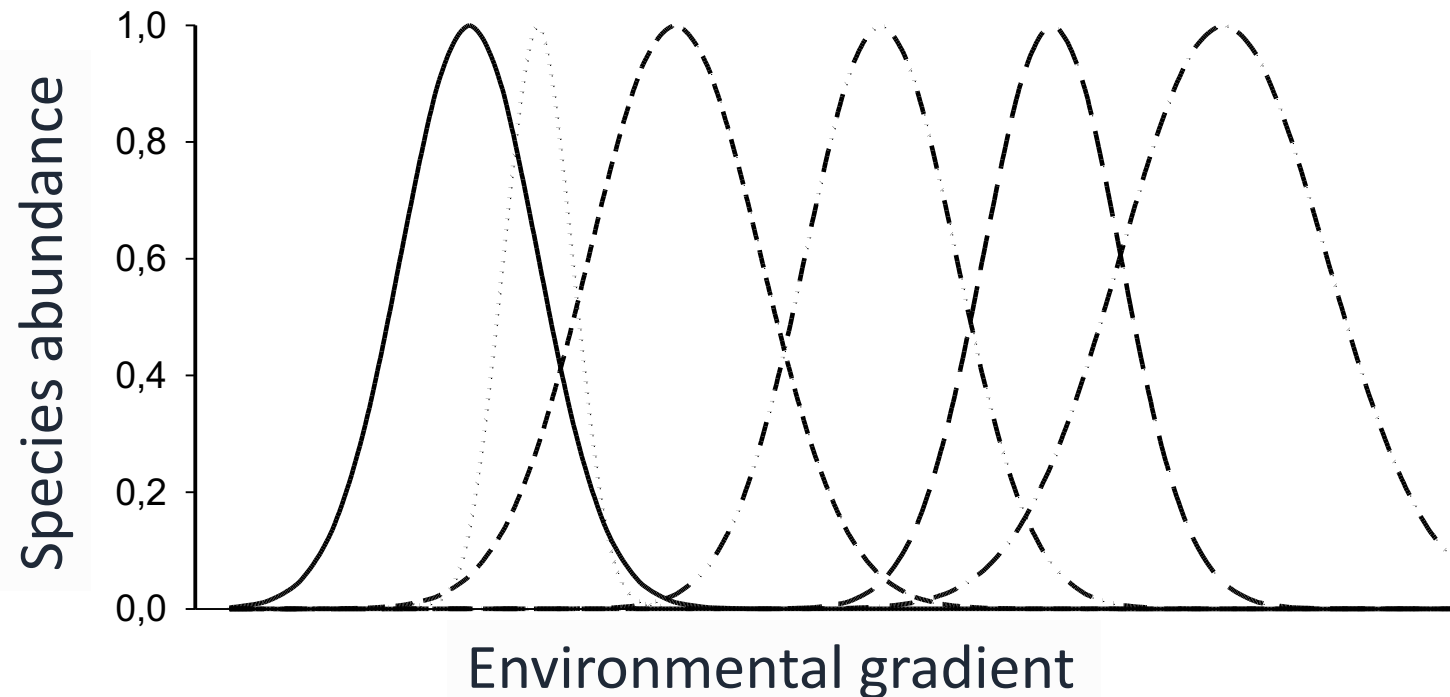


Ordination



Linear $\leftrightarrow$ unimodal Ordination methods?

- Depends on lengths of gradient
- Long gradient $\rightarrow$ unimodal „relation“

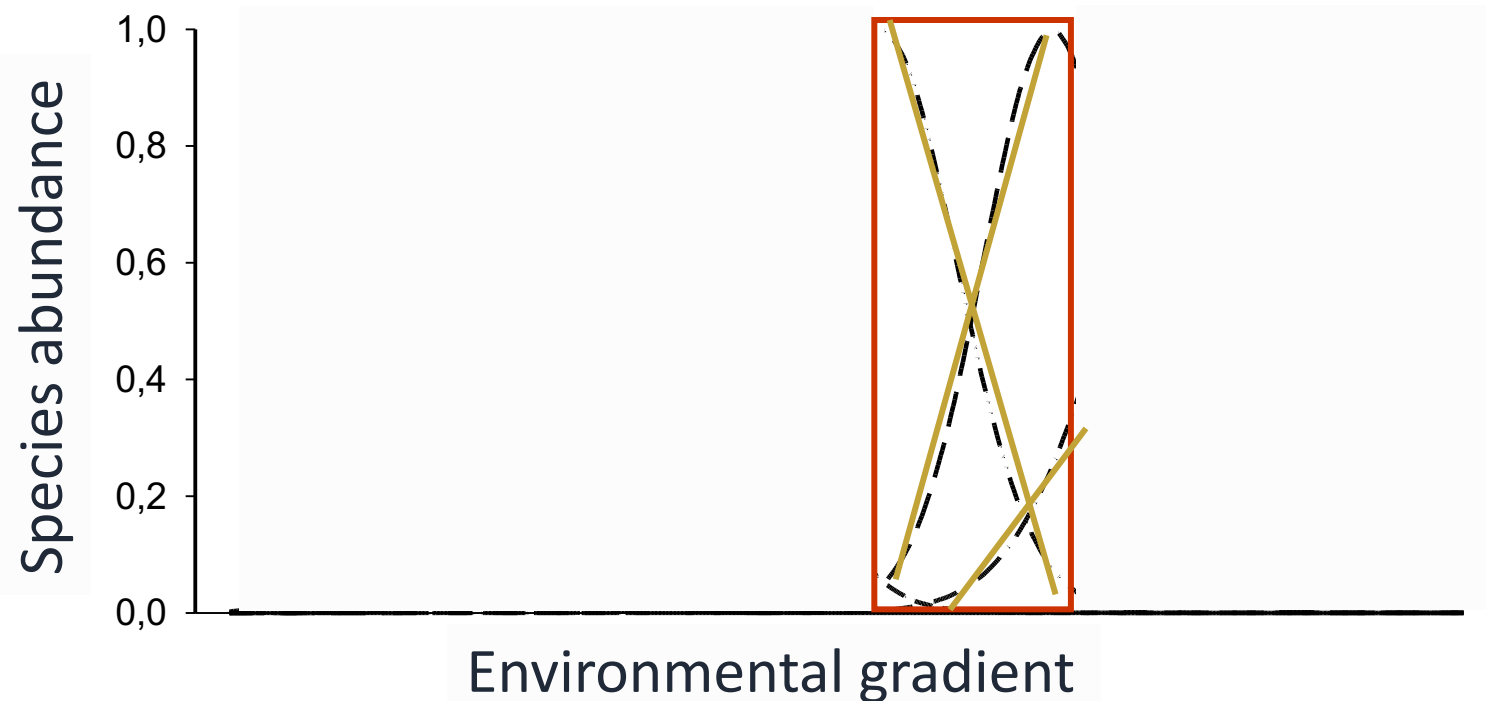


Ordination



Linear $\leftrightarrow$ unimodal Ordination methods?

- Depends on lengths of gradient
- Long gradient $\rightarrow$ unimodal „relation“



Overview on ordination methods



	Indirect (no environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis



Overview on ordination methods

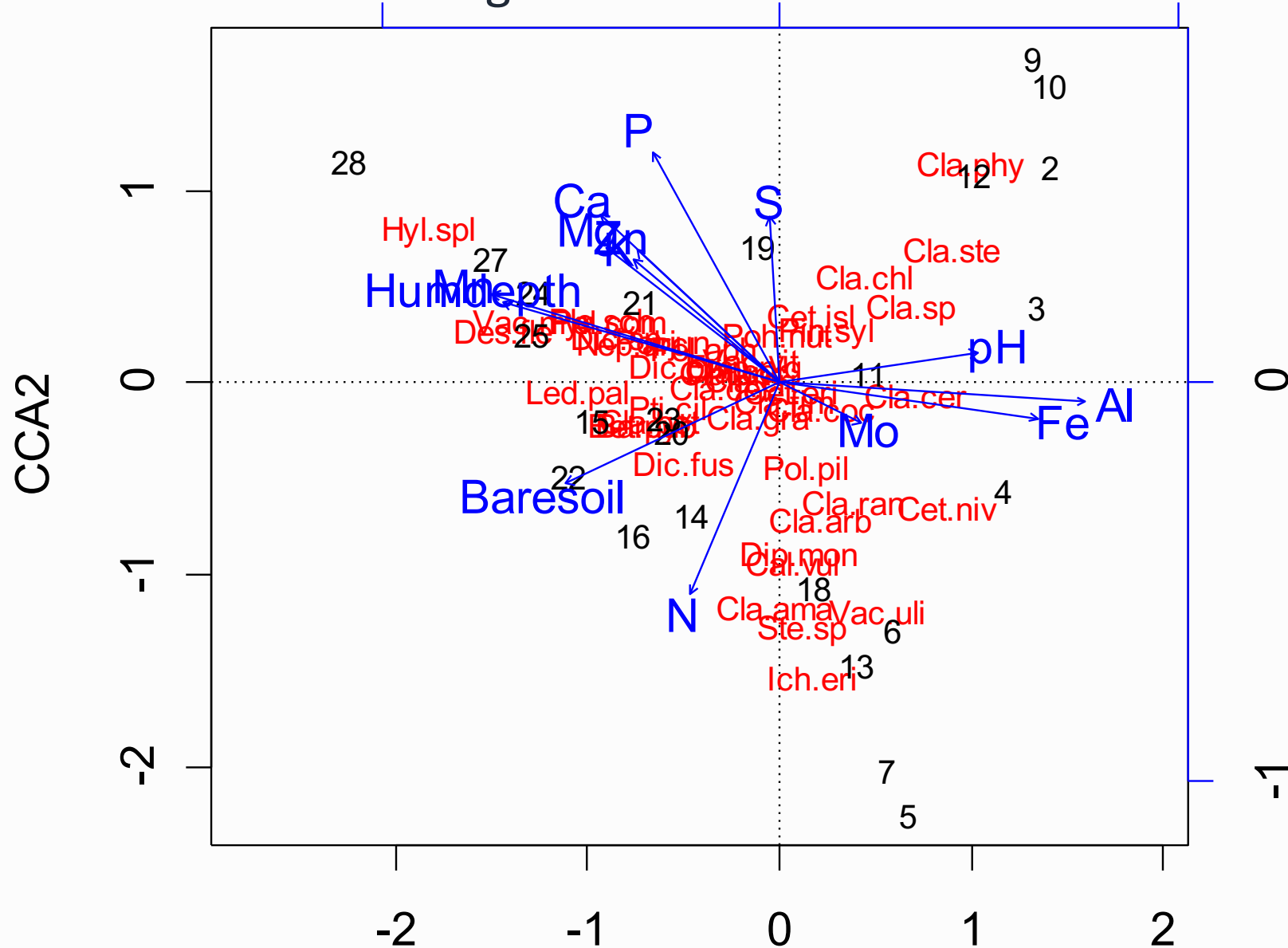
	Indirect (no environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis
Any relation	NMDS (Non-metric multidimensional scaling)

Overview on ordination methods

	Indirect (no environmental data)	Direct (with environmental data)
Linear relation	PCA (PCoA) Principal Components/ Coordinates Analysis	RDA Redundancy Analysis
Unimodel relation	CA (DCA) (Detrended) Correspondence Analysis	CCA Canonical CA
Any relation	NMDS (Non-metric multidimensional scaling)	



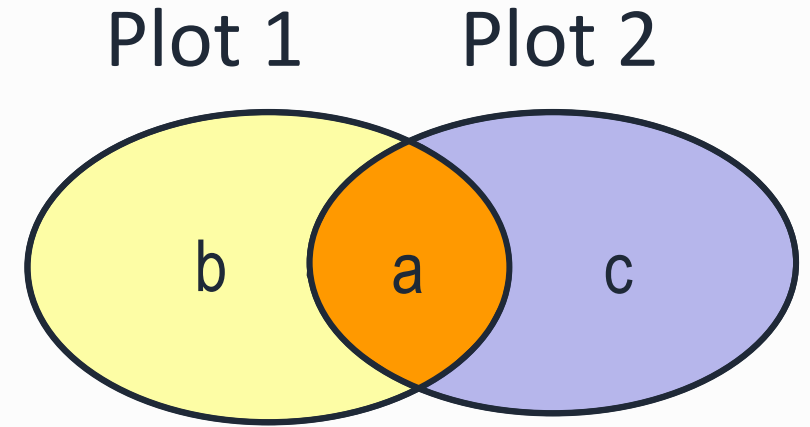
CCA – with all environmental gradients





Similarity- and dissimilarity indices

		<i>Plot 1</i>	
		+	-
<i>Plot 2</i>	+	a	c
	-	b	d



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

e.g.: Sørensen similarity = $2a / (2a+b+c)$

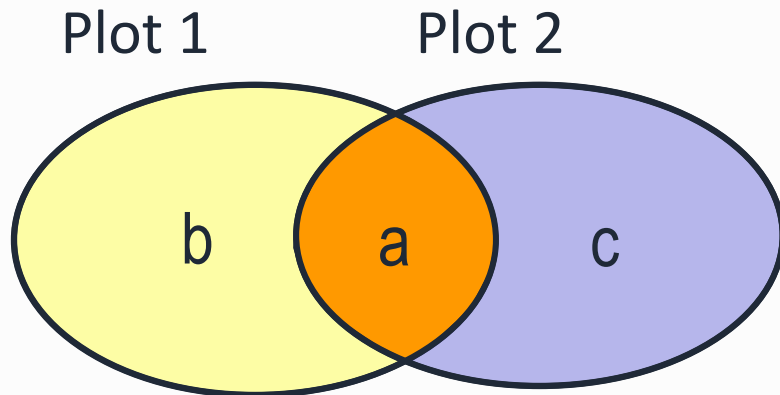
Correspondence Analysis



Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$sim = \frac{(a+b+c+d)(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}$$

Yule & Kendall (1950) in Jurasinski (2012)



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

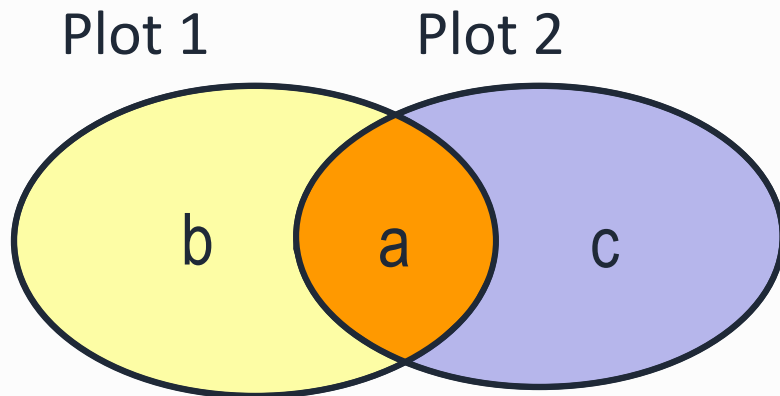
d: Species not in the two plots

Correspondence Analysis



Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

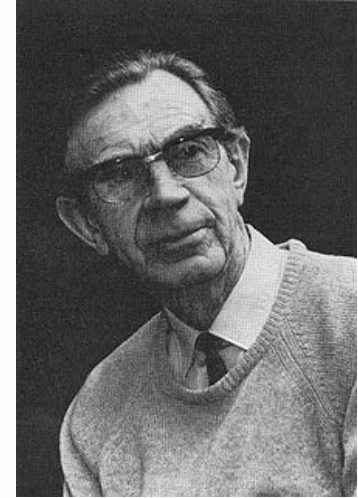
d: Species not in the two plots

Correspondence Analysis

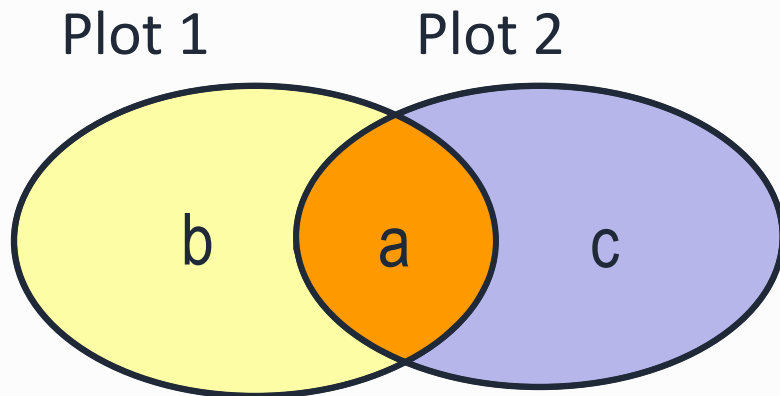


Correspondence Analysis implicitly uses Chi-square distance/similarity. Which can be expressed as:

$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$



Thorvald Sørensen



- a:** Shared species
- b:** Plot 1 exclusive species
- c:** Plot 2 exclusive species
- d:** Species not in the two plots



(Dis)Similarity Matrices

	v_1	v_2	v_3	v_4
o_1	$V_{1,1}$	$V_{1,2}$	$V_{1,3}$	$V_{1,4}$
o_2	$V_{2,1}$	$V_{2,2}$	$V_{2,3}$	$V_{2,4}$
o_3	$V_{3,1}$	$V_{3,2}$	$V_{3,3}$	$V_{3,4}$
o_4	$V_{4,1}$	$V_{4,2}$	$V_{4,3}$	$V_{4,4}$
o_5	$V_{5,1}$	$V_{5,2}$	$V_{5,3}$	$V_{5,4}$
o_6	$V_{6,1}$	$V_{6,2}$	$V_{6,3}$	$V_{6,4}$

Raw data

dissimilarity
matrix

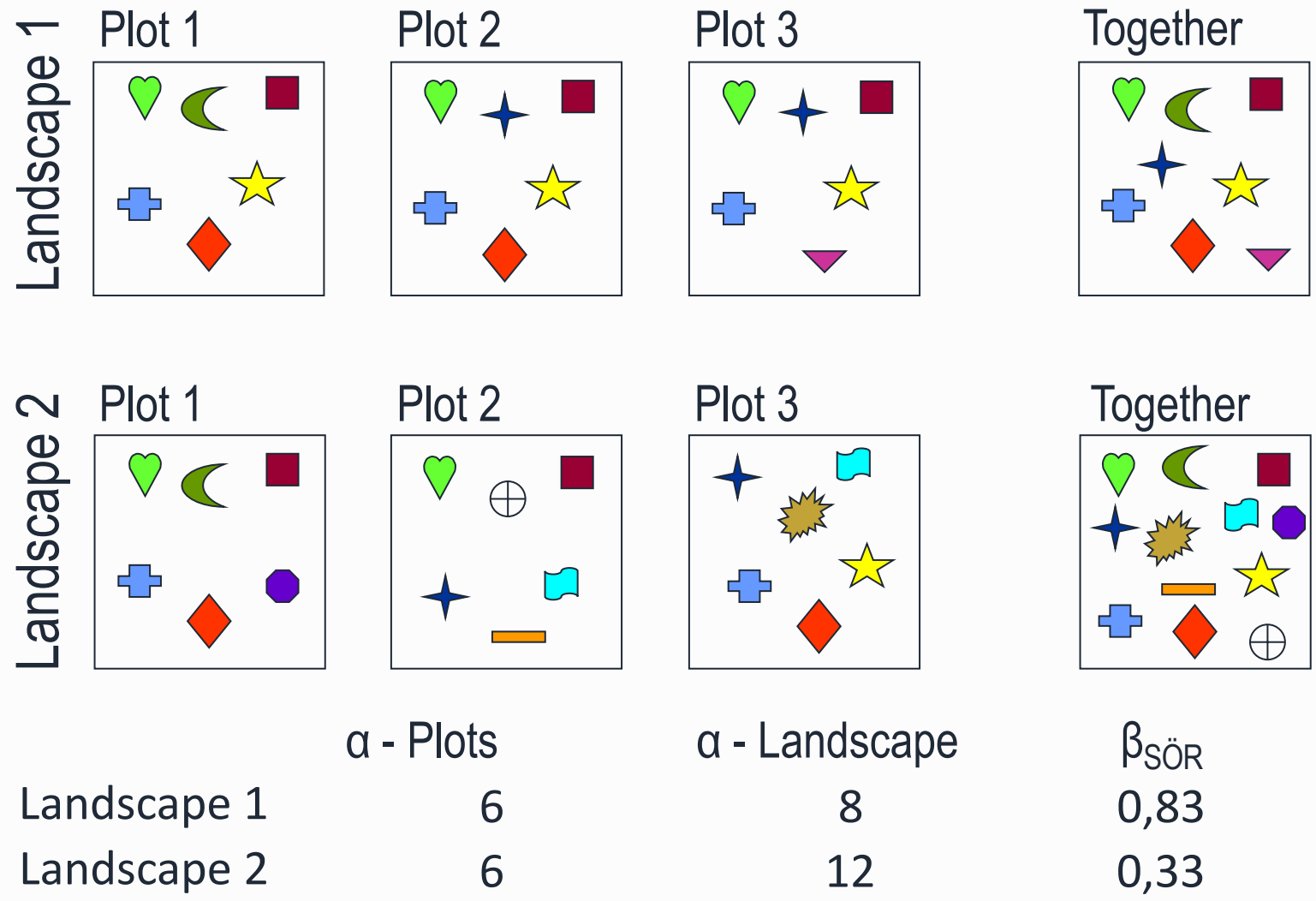


o_1	0					
o_2	$d_{2,1}$	0				
o_3	$d_{3,1}$	$d_{3,2}$	0			
o_4	$d_{4,1}$	$d_{4,2}$	$d_{4,3}$	0		
o_5	$d_{5,1}$	$d_{5,2}$	$d_{5,3}$	$d_{5,4}$	0	
o_6	$d_{6,1}$	$d_{6,2}$	$d_{6,3}$	$d_{6,4}$	$d_{6,5}$	0

similarity
matrix

	O_1	O_2	O_3	O_4	O_5	O_6
O_1	1					
O_2	$S_{2,1}$	1				
O_3	$S_{3,1}$	$S_{3,2}$	1			
O_4	$S_{4,1}$	$S_{4,2}$	$S_{4,3}$	1		
O_5	$S_{5,1}$	$S_{5,2}$	$S_{5,3}$	$S_{5,4}$	1	
O_6	$S_{6,1}$	$S_{6,2}$	$S_{6,3}$	$S_{6,4}$	$S_{6,5}$	1

Why do we need these measures?



Diversity of indices



$$\beta_{\text{Sørensen}} = \frac{2a}{2a + b + c}$$

$$\beta_{\text{Williams}} = \frac{\min(b, c)}{a + b + c}$$

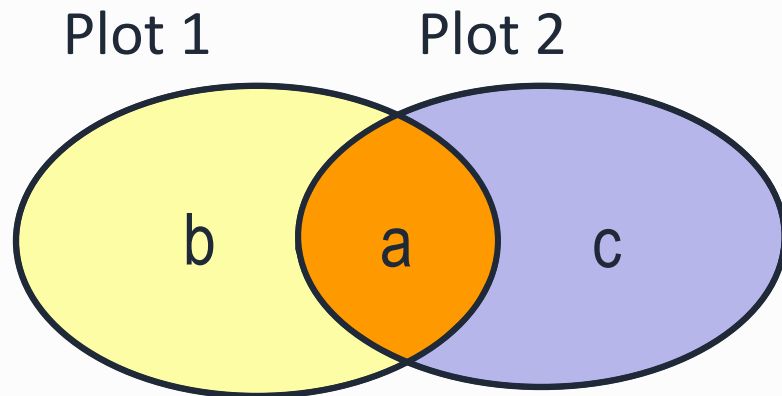
$$\beta_{\text{Jaccard}} = \frac{a}{a + b + c}$$

$$\beta_{\text{Lande}} = \frac{b + c}{2}$$

$$\beta_{\text{Wilson \& Shmida}} = \frac{b + c}{2a + b + c}$$



Diversity of indices



soerensen

$$sim = \frac{2a}{2a+b+c}$$

jaccard

$$sim = \frac{a}{a+b+c}$$

ochiai

$$sim = \frac{a}{\sqrt{(a+b)(a+c)}}$$

mountford

$$sim = \frac{2a}{(a(b+c)+2bc)}$$

whittaker

$$sim = \frac{a+b+c}{2a+b+c} - 1$$

lande

$$sim = \frac{b+c}{2}$$

wilsonshmid

$$sim = \frac{b+c}{2a+b+c}$$

cocogaston

$$sim = \frac{b+c}{a+b+c}$$

magurran

$$sim = (2a+b+c)(1 - \frac{a}{a+b+c})$$

harrison

$$sim = \frac{\min(b,c)}{\max(b,c)+a}$$

cody

$$sim = 1 - \frac{a(2a+b+c)}{2(a+b)(a+c)}$$

williams

$$sim = \frac{\min(b,c)}{a+b+c}$$

williams2

$$\frac{(bc)+1}{(a+b+c)^2 - (a+b+c)}$$

harte

$$1 - \frac{2a}{2a+b+c}$$

simpson

$$\frac{\min(b,c)}{\min(b,c)+a}$$

lennon

$$\frac{2|b-c|}{2a+b+c}$$

weiher

$$sim = b + c$$

ruggiero

$$sim = \frac{a}{a+c}$$

lennon2

$$sim = 1 - \left[\frac{\log\left(\frac{2a+b+c}{a+b+c}\right)}{\log 2} \right]$$

routledge

$$sim = \frac{(a+b+c)^2}{(a+b+c)^2 - 2bc}$$

routledge

toolong, see below

routledge

$$sim = e^{routledge} - 1$$

sokal1

$$sim = \frac{a}{a+2(b+c)}$$

dice

$$sim = \frac{a}{\min((b+a), (c+a))}$$

kulczlinsky

$$sim = \frac{a}{b+c}$$

Soerensen (1948)

Jaccard (1912)

Ochiai (1957), Shi (1993)

Mountford (1962), Shi (1993)

Whittaker (1960), Magurran (1988)

Lande (1996)

Wilson & Shmida (1984)

Colwell & Coddington (1948), Gaston et al. (2001)

Magurran (1988)

Harrison et al. (1992), Koleff et al. (2003)

Cody (1993)

Williams (1996), Koleff et al. (2003)

Williams (1996), Koleff et al. (2003)

Harte & Kinzig (1997), Koleff et al. (2003)

Simpson (1949), Koleff et al. (2003)

Lennon et al. (2001), Koleff et al. (2003)

Weiher & Boylen (1994)

Ruggiero et al. (1998), Koleff et al. (2003)

Lennon et al. (2001), Koleff et al. (2003)

Routledge (1977), Magurran (1988)

Routledge (1977), Wilson & Shmida (1984)

Routledge (1977)

Sokal & Sneath (1963)

Association index of Dice (1945), Wolda (1981)

Oosting (1956), Southwood (1978)

Diversity of indices



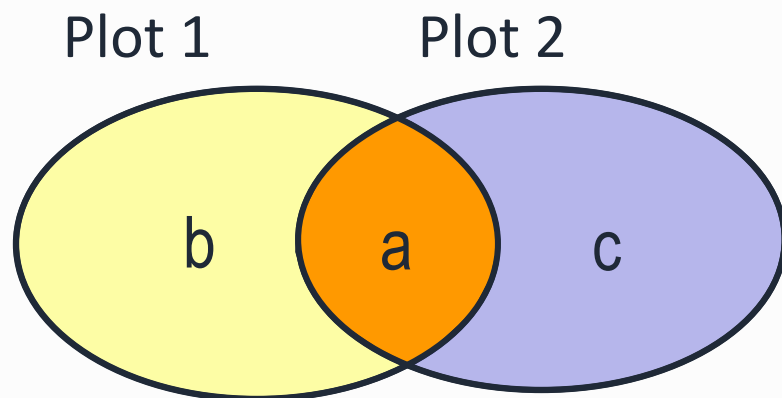
kulcz2insky	$sim = \frac{a}{(a+b)(a+c)}$
mcconnagh	$sim = \frac{a^2 - bc}{(a+b)(a+c)}$
simpson2	$sim = \frac{a}{a+b}$
legendre2	$sim = \frac{3a}{3a+b+c}$
fager	$sim = \frac{a}{\sqrt{n_1 n_2}} - \frac{1}{2\sqrt{n_2}}$
maarel	$sim = \frac{2a - (b+c)}{2a+b+c}$
lamont	$sim = \frac{a}{2a+b+c}$
johnson	$sim = \frac{a}{2b}$
sorgenfrei	$sim = \frac{a^2}{(a+b)(a+c)}$
johnson2	$sim = \frac{a}{a+b} + \frac{a}{a+c}$

Oosting (1956), Southwood (1978)
 Hubalek (1982)
 Simpson (1960), Shi (1993)
 Legendre & Legendre (1998)
 Fager (1957), Shi (1993)
 van der Maarel (1969)
 Lamont and Grant (1979)
 Johnson (1971)
 Sorgenfrei (1959)
 Johnson (1967)

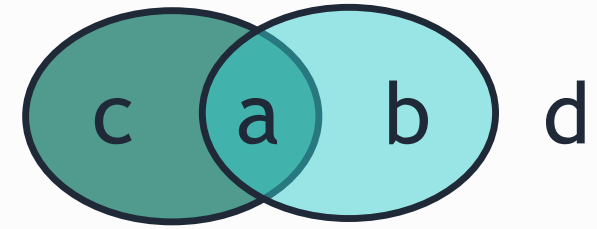
Computable symmetric indices (including unshared species):

manhattan	$sim = \frac{b+c}{a+b+c+d}$
simplematching	$sim = \frac{a+d}{a+b+c+d}$
margaleff	$sim = \frac{a(a+b+c+d)}{(a+b)(a+c)}$
pearson	$sim = \frac{ad-bc}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}$
roger	$sim = \frac{a+d}{a+2(b+c)+d}$
baroni	$sim = \frac{\sqrt{ad+c}}{\sqrt{ad+a+b+c}}$
dennis	$sim = \frac{ad-bc}{\sqrt{(a+b+c+d)(a+b)(a+c)}}$
fossum	$sim = \frac{(a+b+c+d)(-\frac{a}{2})^2}{(a+b)(a+c)}$
gower	$sim = \frac{a-(b+c)+d}{a+b+c+d}$
legendre	$sim = \frac{a}{a+b+c+d}$
sokal2	$sim = \frac{ad}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}$
sokal3	$sim = \frac{2a+2d}{(a+d)+(a+b+c+d)}$
sokal4	$sim = \frac{\frac{a+d}{b+c}}{(a+b+c+d)(\frac{ ad-bc - \frac{a+b+c+d}{2}}{2})^2}$
stiles	$sim = \frac{ad-bc}{ad+bc}$
yule	$sim = \frac{4(ad-bc)}{(a+b)^2 + (b+c)^2}$
michael	$sim = \frac{(a+d) - (b+c)}{(a+d) - (b+c)}$
hamann	$sim = \frac{N}{(aN-2n_2)}$
forbes	$sim = \frac{2(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)}$
chisquare	$sim = \frac{(ad-bc)}{(a+c)(b+d)}$
peirce	$sim = \frac{a-(a+b)(a+c)}{(a+b)(a+c)(b+d)(c+d)}$
eyraud	$sim = \frac{a-(a+b)(a+c)}{(a+b)(a+c)(b+d)(c+d)}$

Mean Manhattan, Legendre & Legendre (1998)
 Sokal & Michener 1958
 Clifford & Stevenson (1975)
 Phi of Pearson, Gower & Legendre (1986)
 Rogers & Tanimoto (1960), Gower & Legendre (1986)
 Baroni-Urbani & Buser (1976), Wolda (1981)
 Holliday et al. (2002)
 Holliday et al. (2002)
 Gower & Legendre (1986)
 Gower & Legendre (1986)
 Sokal & Sneath (1963)
 Sokal & Sneath (1963)
 Sokal & Sneath (1963)
 Stiles (1946)
 Yule & Kendall (1973)
 Michael (1920), Shi (1993)
 Hamann (1961)
 Forbes (1925), Shi (1993)
 Yule & Kendall (1950)
 Peirce (1984) in Shi (1993)
 Eyraud (1936) in Shi (1993)



Diversity of indices



$$\beta_{ij} = (\varepsilon a + \delta d) (\varepsilon a + \delta d + \lambda(b + c))^{-1}$$

β_{ij} = Similarity Value for Data Sets i, j

a = Species common in i and j

b = Species only in i

c = Species only in j

d = Species neither in i nor in j but in the area

$\varepsilon, \delta, \lambda$ = coefficients

Sørensen's Coefficient is binary



		Object 1	
		1	0
Object 2	1	a	b
	0	c	d

$$S = \frac{2a}{2a + b + c}$$

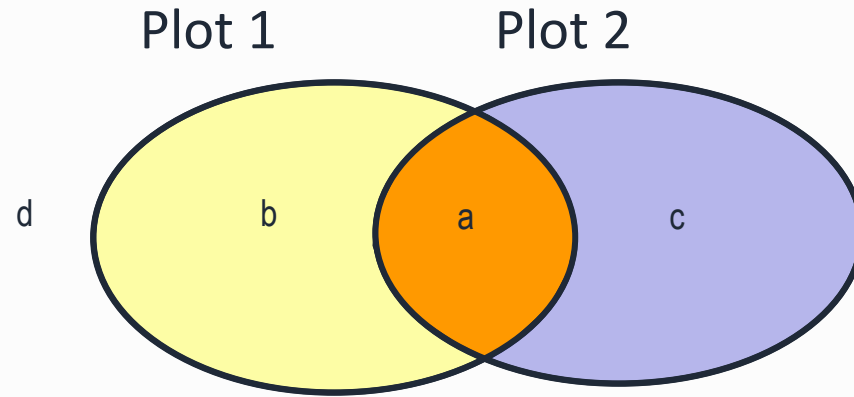
Bray-Curtis (=Steinhaus' Coefficient)

Bray-Curtis is Sørensen coefficient but including abundance information:

$$BC_{ij} = \frac{2C_{ij}}{S_i + S_j}$$

C = sum of the lesser abundance values for shared species
S = total abundance of species counted at both sites

Similarity- and dissimilarity indices



a: Shared species

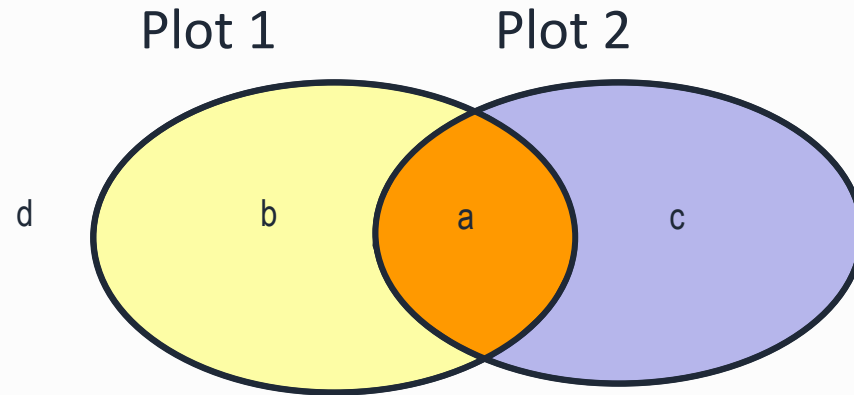
b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Further species



Similarity- and dissimilarity indices



a: Shared species

b: Plot 1 exclusive species

c: Plot 2 exclusive species

d: Further species

e.g.: Sørensen similarity = $2a / (2a+b+c)$

	V_1	V_2	V_3	V_4
O_1	$V_{1,1}$	$V_{1,2}$	$V_{1,3}$	$V_{1,4}$
O_2	$V_{2,1}$	$V_{2,2}$	$V_{2,3}$	$V_{2,4}$
O_3	$V_{3,1}$	$V_{3,2}$	$V_{3,3}$	$V_{3,4}$
O_4	$V_{4,1}$	$V_{4,2}$	$V_{4,3}$	$V_{4,4}$
O_5	$V_{5,1}$	$V_{5,2}$	$V_{5,3}$	$V_{5,4}$
O_6	$V_{6,1}$	$V_{6,2}$	$V_{6,3}$	$V_{6,4}$

Raw data



similarity
matrix

	O_1	O_2	O_3	O_4	O_5	O_6
O_1	1					
O_2	$S_{2,1}$	1				
O_3	$S_{3,1}$	$S_{3,2}$	1			
O_4	$S_{4,1}$	$S_{4,2}$	$S_{4,3}$	1		
O_5	$S_{5,1}$	$S_{5,2}$	$S_{5,3}$	$S_{5,4}$	1	
O_6	$S_{6,1}$	$S_{6,2}$	$S_{6,3}$	$S_{6,4}$	$S_{6,5}$	1



	O_1	O_2	O_3	O_4	O_5	O_6
O_1	0					
O_2	$d_{2,1}$	0				
O_3	$d_{3,1}$	$d_{3,2}$	0			
O_4	$d_{4,1}$	$d_{4,2}$	$d_{4,3}$	0		
O_5	$d_{5,1}$	$d_{5,2}$	$d_{5,3}$	$d_{5,4}$	0	
O_6	$d_{6,1}$	$d_{6,2}$	$d_{6,3}$	$d_{6,4}$	$d_{6,5}$	0

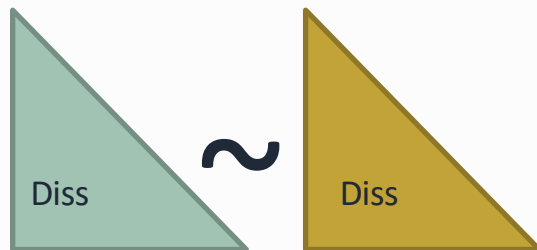
Dissimilarity matrix



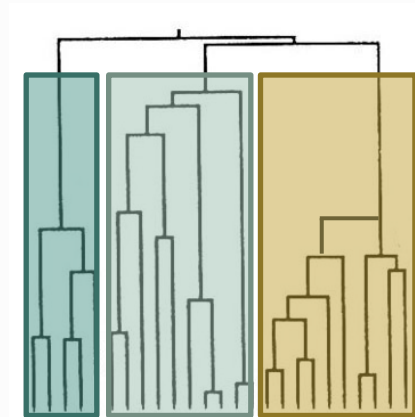
or



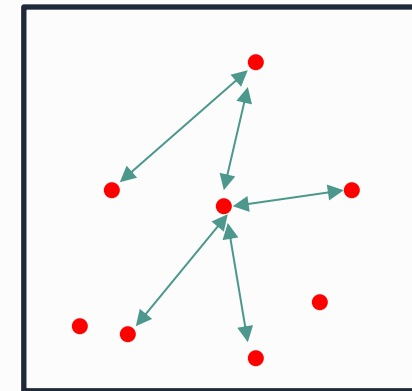
or



Relate to other
dissimilarity matrix



Classification

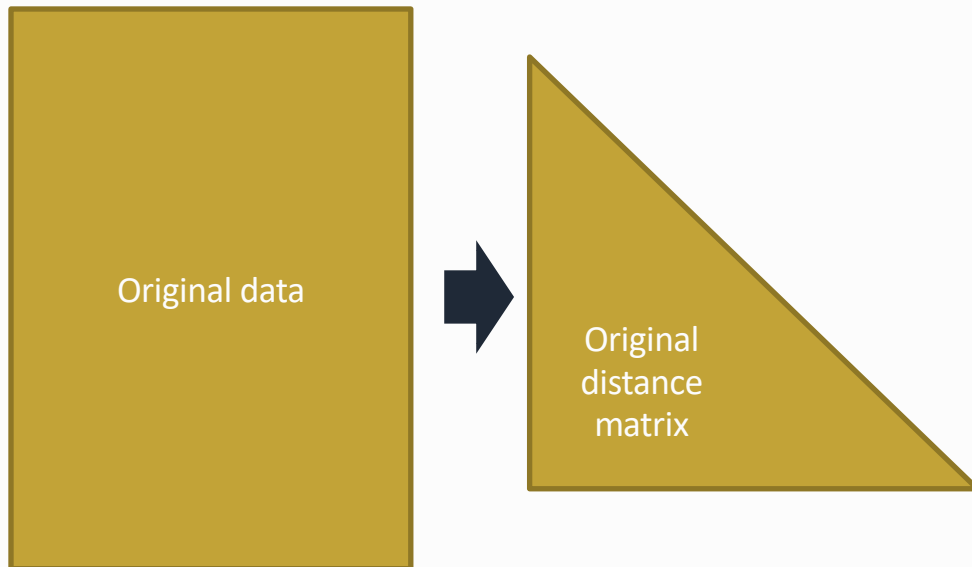


Dimension reduction
(e.g. NMDS)



(N)MDS - The goal

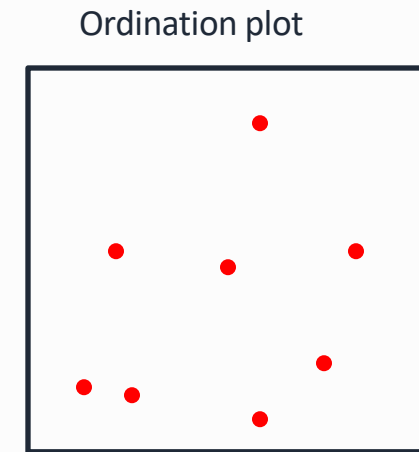
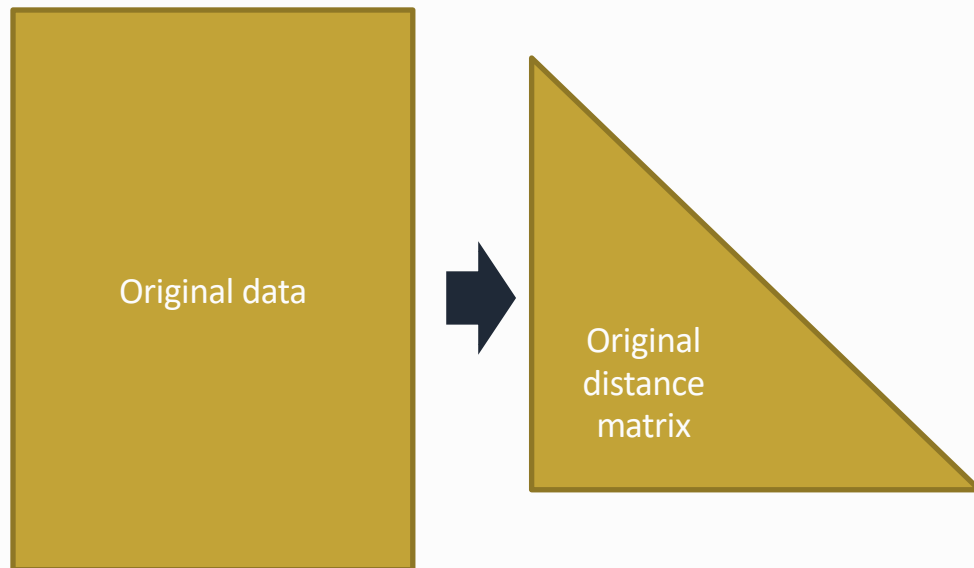
- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





(N)MDS - The goal

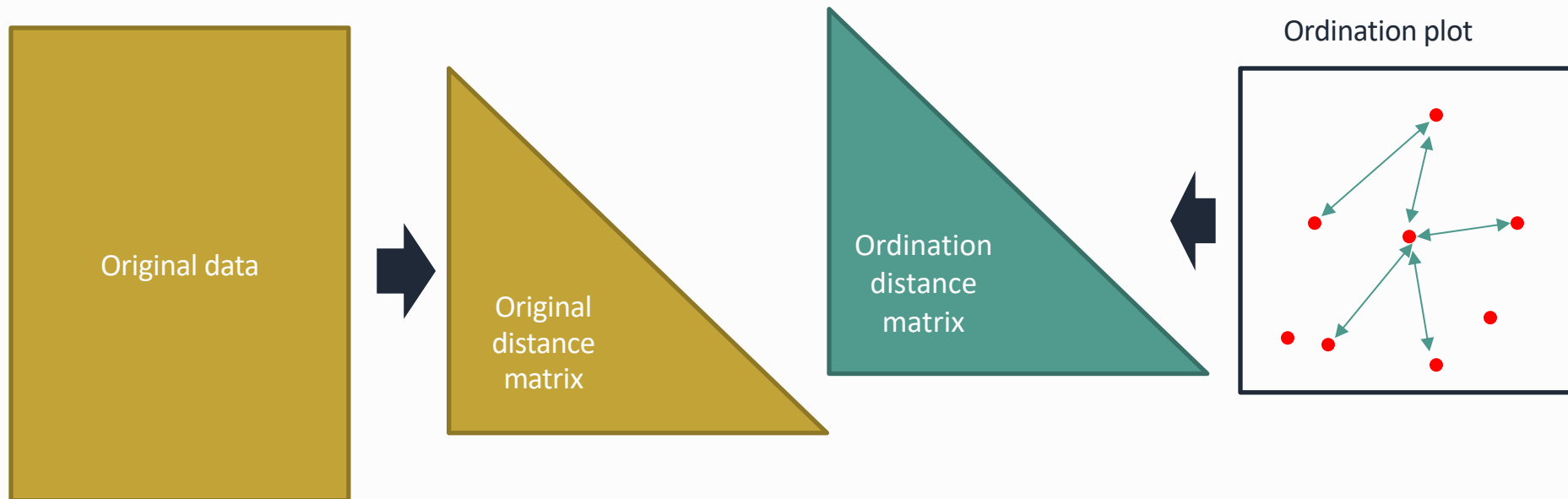
- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





(N)MDS - The goal

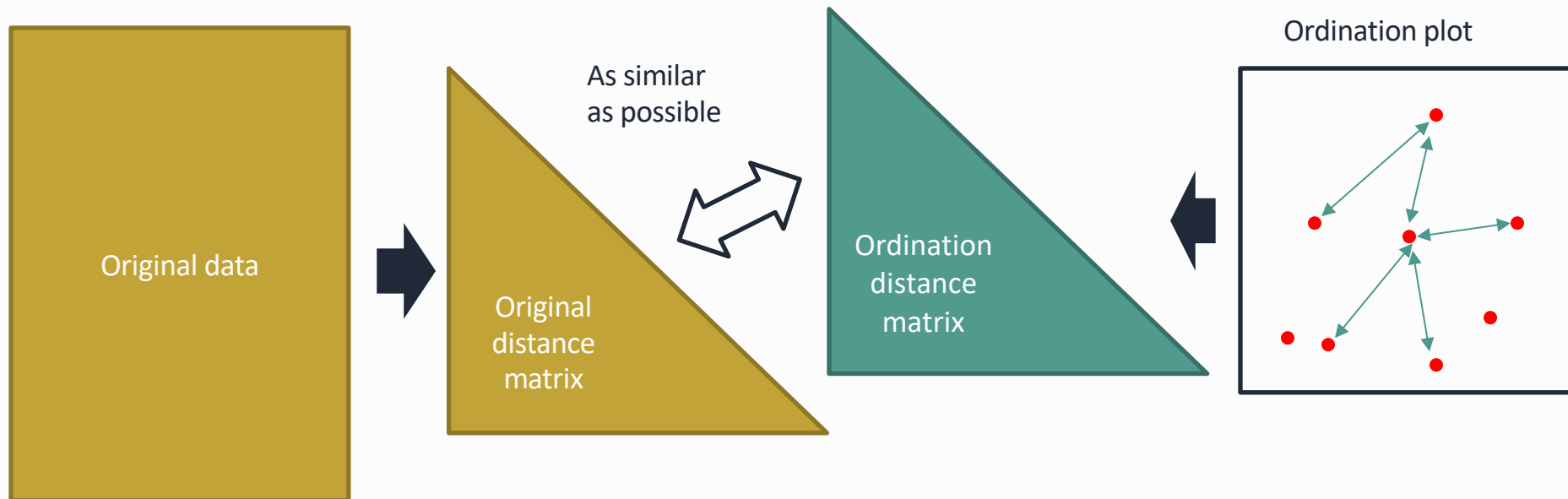
- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





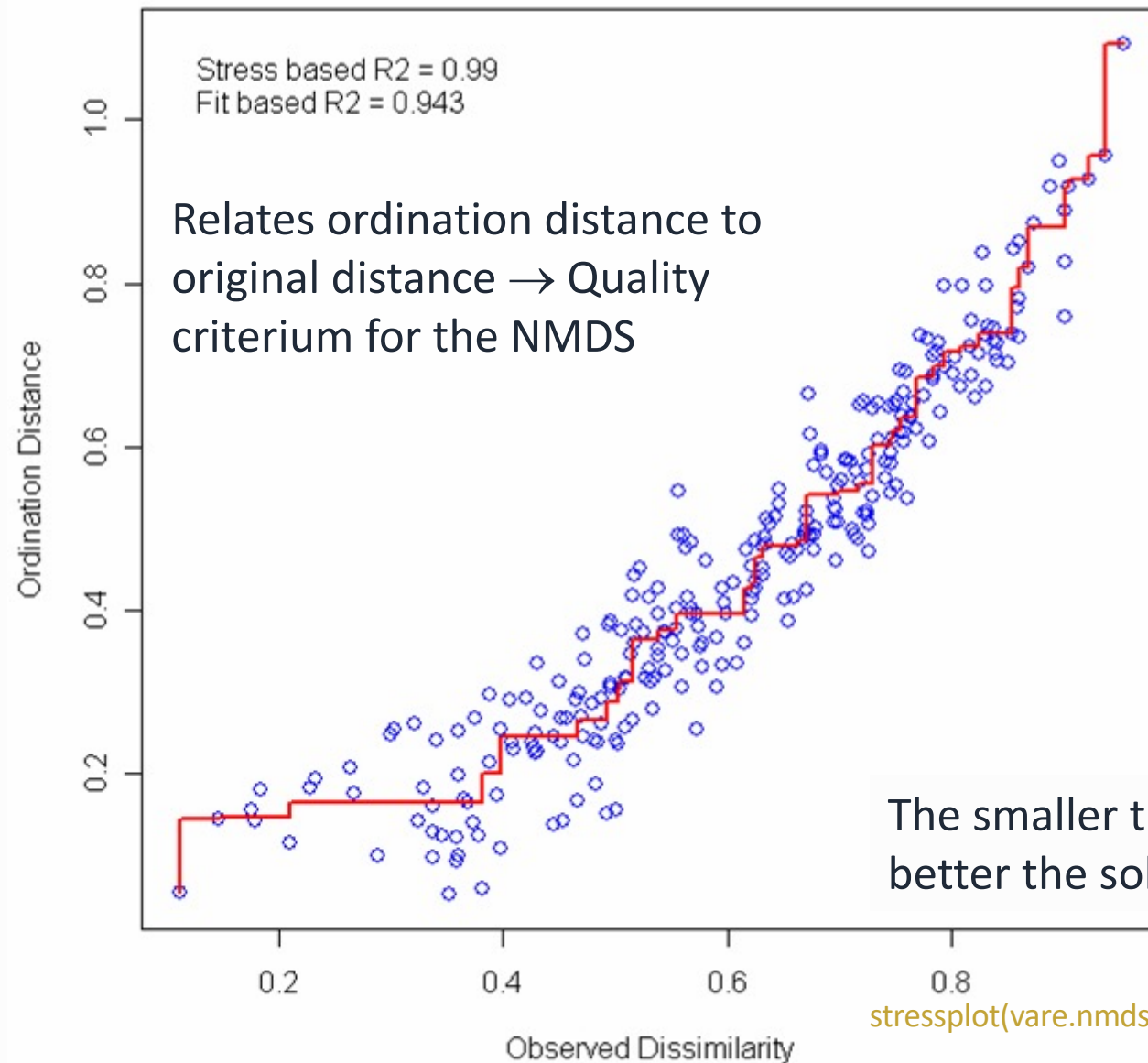
(N)MDS - The goal

- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





NMDS – stressplot

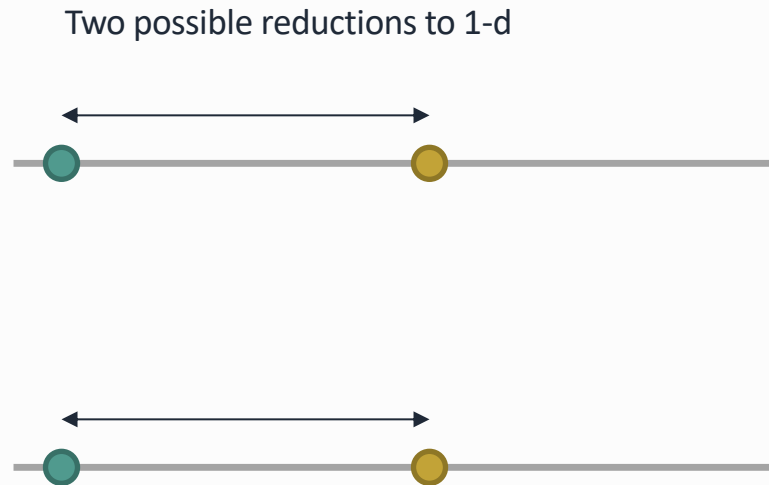
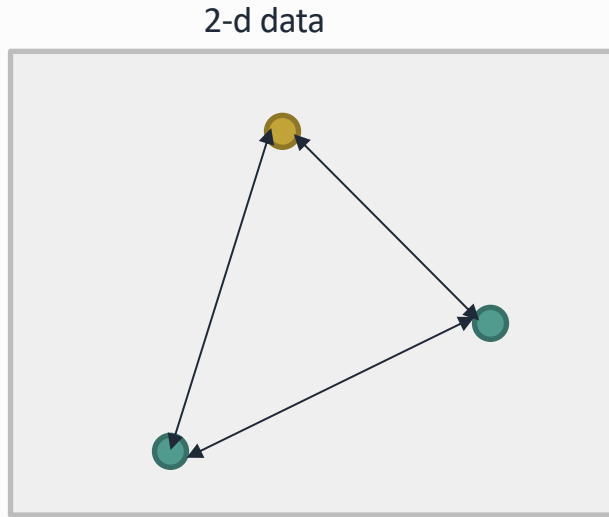


Ordination



Distortion:

- Reducing high dimensional data to low dimensional data causes loss of information:

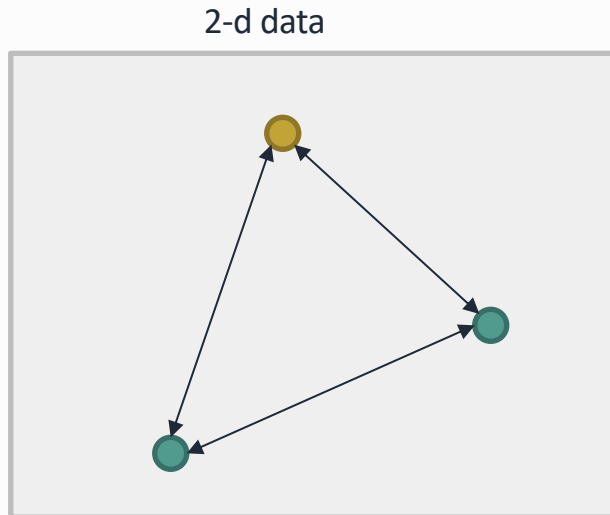


Ordination



Distortion:

- Reducing high dimensional data to low dimensional data causes loss of information:

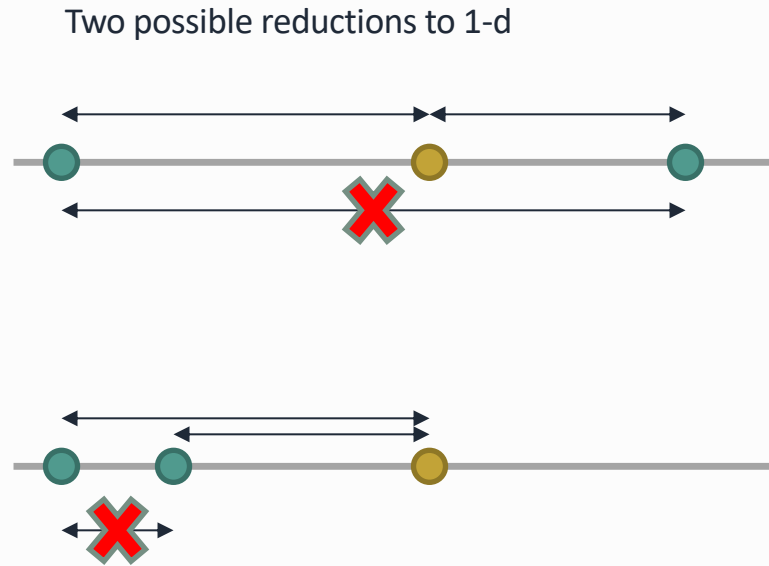
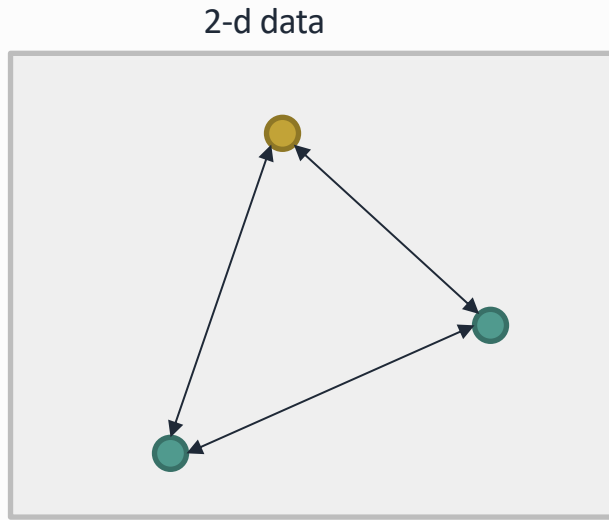


Ordination



Distortion:

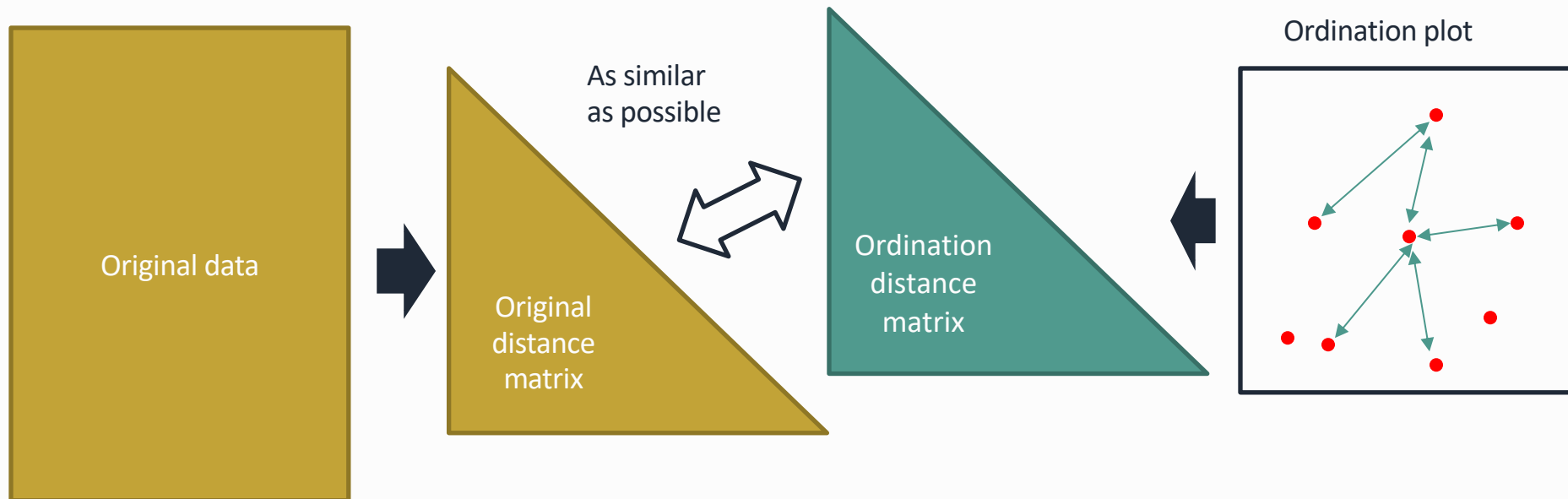
- Reducing high dimensional data to low dimensional data causes loss of information:





(N)MDS

- Visualization!
- Create 2d or 3d plot that tries to show the original distances as well as possible





	O_1	O_2	O_3	O_4	O_5	O_6
O_1	0					
O_2	$d_{2,1}$	0				
O_3	$d_{3,1}$	$d_{3,2}$	0			
O_4	$d_{4,1}$	$d_{4,2}$	$d_{4,3}$	0		
O_5	$d_{5,1}$	$d_{5,2}$	$d_{5,3}$	$d_{5,4}$	0	
O_6	$d_{6,1}$	$d_{6,2}$	$d_{6,3}$	$d_{6,4}$	$d_{6,5}$	0

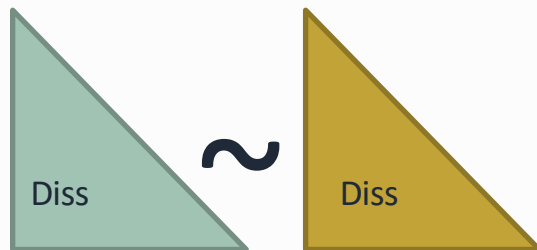
Dissimilarity matrix



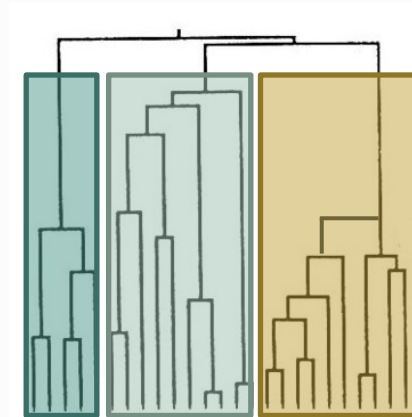
or



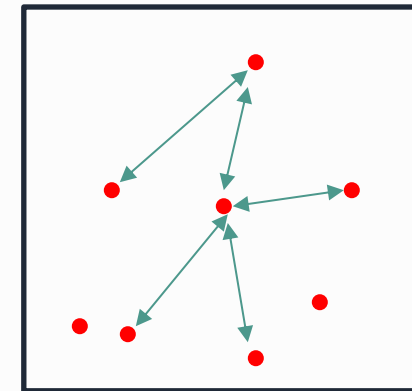
or



Relate to other
dissimilarity matrix



Classification



Dimension reduction
(e.g. NMDS)

Questions?





Tutorials

- Great web-page with resources on Community Data in R by David Zelený:
www.davidzeleny.net/anadat-r/doku.php/en:start

Thank you!